

Critical Replication of Shaikh’s Capacity Utilization Measure*

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Abstract

Identifying a productive capacity ceiling requires estimating an unobserved benchmark, a task that risks conflating technical engineering limits with the institutional realities of capitalist accumulation. This chapter critically replicates Anwar Shaikh’s accounting-based capacity measure for the post-war US corporate sector (1947–2011), reinterpreting the long-run output–capital coefficient as a capacity transformation elasticity (θ) under conditions of unbalanced growth. Testing this relation across a combinatorial grid of 500 single-equation specifications and a system-level Vector Error Correction Model (VECM), we demonstrate that the standalone bivariate output-capital relation fails to cointegrate. System-level stability emerges exclusively when the rate of exploitation (e_t) enters the cointegrating vector, yielding an estimated transformation elasticity ($\hat{\theta} < 1$) indicative of structural overaccumulation. This evidence suggests that the productive ceiling is not a neutral engineering constraint. Rather, it is a contingent outcome of decentralized accumulation, structurally conditioned by the distributive conflict and the labor process within and beyond the workplace, which physically dictate the conversion of accumulated capital into active productive capacity.

Keywords: Capacity Utilization; Critical Replication; Shaikh; Transformation Elasticity; Cointegration; ARDL; VECM; Distribution.

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1 Introduction

1.1 The unobserved productive ceiling is not a neutral engineering constraint; it is the central macroeconomic variable linking capital accumulation, profitability, and class struggle. While actual output is driven by effective demand, productive capacity reflects the institutional limits of exploitation and the physical intensity of the labor process. [Shaikh \(2016\)](#) offers a portable, accounting-based solution to measure this ceiling via the long-run relation between aggregate output and capital stock. However, treating the estimated output–capital coefficient as a stable, physical law of production risks masking the historical and distributive conflicts that govern the workplace. This chapter critically replicates and extends Shaikh’s strategy to determine whether this empirical coefficient represents a stable technical relation or a historically contingent reflection of class struggle.

1.2 Distinct from the baseline approach of [Shaikh \(2016\)](#), this chapter reframes the aggregate output–capital coefficient as a capacity transformation elasticity (θ). We define the latent capacity ceiling in log-linear form as:

$$\ln Y_t^p = \theta \ln K_t + \ln R_0^n, \quad (1)$$

where Y_t^p is potential capacity, K_t is the capital stock, and R_0^n is a scale constant. Under the standard balanced-growth assumption of post-Keynesian models, this elasticity is constrained to unity ($\theta = 1$), meaning capacity expands in lockstep with capital. If $\theta \neq 1$, however, capacity formation decouples from accumulation. An elasticity below unity ($\theta < 1$) characterizes a structural overaccumulation regime where capital growth outpaces capacity expansion, while $\theta > 1$ leads to chronic excess capacity. By analyzing θ directly, we move beyond Shaikh’s focus on short-run residual fluctuations to investigate the long-run structural stability of the output–capital relation.

1.3 Evaluating this relationship reveals that Shaikh’s baseline results are approximately recoverable, yet the underlying elasticity is highly sensitive to model selection. In replicating the single-equation ARDL model of [Shaikh \(2016\)](#) using US corporate sector data (1947–2011), we come close to but do not exactly reproduce the original estimates due to undocumented lag structures and price deflator choices. Interrogating this fragility across a combinatoric specification space of 500 ARDL models reveals “disciplined non-uniqueness.” Rather than converging to a single technical parameter, the estimated elasticity ranges from 0.65 to 0.95. This parameter volatility demonstrates that the recovered capacity path is a statistical artifact of researcher-chosen lag lengths and historical dummy variables, rather than a unique physical law.

1.4 Shifting the analysis to a joint system framework reveals that the bivariate output–capital relationship is spurious. In a bivariate Vector Error Correction Model (VECM), the output–capital relation fails cointegration and dynamic stability tests across all lag profiles. Cointegration and system-level stability emerge only when the logged rate of exploitation (e_t) enters the state vector in a trivariate VECM. This cointegrating survival provides the central empirical finding of the chapter: the rate of exploitation is not a neutral accounting ratio, but a proxy for the balance of class power that dictates shop-floor discipline, machine speed-ups, and the choice of technique. The empirical requirement of e_t in the cointegrating vector indicates that capacity utilization is a distributionally conditioned phenomenon, requiring explicit class struggle controls to establish a stable long-run attractor.

1.5 These findings define the historical limits of the accounting-based capacity measurement of [Shaikh \(2016\)](#). While the output–capital ratio remains a useful anchor to construct capacity utilization, the resulting ceiling cannot be treated as an insulated engineering benchmark. Cointegration survives only when the model is conditioned by the rate of exploitation and stabilized by historical dummy controls that mark structural breaks in post-war US accumulation. Ultimately, the long-run output–capital relation cannot be estimated in isolation from the institutional conflicts of the capitalist labor process, which physically govern how capital is converted into potential output.

1.6 To establish this argument, the chapter unfolds along a narrative that historicizes the measurement problem before executing the econometric tests. We first reconstruct the historical evolution of capacity metrics in Section 2, tracing how survey indicators and output-gap routines shifted from tracking industrial output to enforcing price-level and wage discipline. In Section 3, we develop the conceptual Leontief framework, showing how omitting distribution introduces an identity trap that biases the capital coefficient, and detailing how [Basu \(2022\)](#)’s viability condition governs endogenous technical change. Section 4 presents the staged empirical stress tests—reconstructing the baseline, mapping the ARDL specification space, and testing system-level VECM cointegration. Section 5 concludes by detailing the theoretical implications of structural overaccumulation ($\hat{\theta} < 1.0$) for profit rate crisis theory, evaluating the interpretation of these distributive coefficients, and outlining the Chapter 2 point of departure for Global South economies.

2 A Historical Trace of Capacity Utilization Measurement

2.1 This section traces the capacity utilization index not as a neutral technical indicator, but as a historical product of macroeconomic statecraft and contested political economy. The metric emerged to make idle capital legible for Keynesian demand management, transformed into a sentinel for inflation control during the crisis of Fordism, and now structures neoliberal policy through technocratic output-gap benchmarks. Throughout these historical phases, the unobserved productive ceiling has remained conceptually underidentified. We trace this trajectory through four steps: the bureaucratic standardization of survey measures, the transition to output-gap governance, the theoretical debates within the post-Keynesian tradition between Sraffian and Kaleckian economists over long-run utilization, and Shaikh’s accounting-based framework.

2.1 From Survey Measures to an Inflation Sentinel: Capacity Utilization during the Fordist Era

2.2 Before capacity utilization could serve as a policy target, early researchers had to resolve a fundamental contradiction in its definition. The mass idleness of the Great Depression forced the state to quantify unused resources, exposing a tension between engineering and economic definitions of productive capacity ([Burns, 1954](#); [Klein and David, 1958](#)). Engineering capacity represents the absolute physical limit of machine operation. Economic capacity, by contrast, defines the profit-maximizing level of output given prevailing wages, input costs, machine speeds, and other factors of technological development. From its inception, measuring capacity was not a neutral descriptive exercise, but an act of statecraft designed to identify the normal conditions of operation for capitalist firms, shaping economic understanding amidst the rise of American Keynesianism.

2.3 During the post-war expansion, Alvin Hansen’s secular stagnation thesis warned that chronic underutilization of capital was an inherent drag on long-run growth, forcing the state to confront the productive ceiling directly (Hansen, 1936, 1941; Barber, 1987). As John Hicks and Paul Samuelson codified these concerns into the IS-LM synthesis, demand management became the core paradigm of macroeconomic policy (Hicks, 1937; Samuelson, 1947, 1948). To stabilize this post-war compromise, the state apparatus required quantitative indicators to detect productive bottlenecks and effective-demand shortfalls. Capacity utilization emerged as the primary metric to render capital idleness legible, transforming abstract stagnation concerns into concrete policy triggers.

2.4 The McGraw-Hill plant-and-equipment surveys standardized this legibility by asking managers to report their operating rates relative to preferred capacity benchmarks. Emerging from the private sector, these surveys served as a central information infrastructure for corporate planning and monopoly capital business strategy, facilitating coordination and forecasting before public initiatives took root (Baran and Sweezy, 1988; Munroe, 2007). The surveys folded heterogeneous, firm-level engineering realities into a uniform statistical reporting stream (Butler, 1958; Phillips, 1963). This private institutionalization depoliticized the capacity concept, replacing structural debates over the efficiency of capitalist production with a standardized language of business forecasting. When the Federal Reserve Board subsequently integrated these survey responses into its official Industrial Production index, it institutionalized these subjective manager estimates as the default benchmark for macroeconomic policy (Board of Governors of the Federal Reserve System, 2017). Alternative, objective physical measures—such as the workweek of capital—never secured comparable bureaucratic adoption (Foss, 1963; Taubman and Gottschalk, 1971).

2.5 The stagflation crisis of the 1970s dismantled the post-war demand-stabilization framework. Faced with falling profit rates and rising inflation, policymakers repurposed the capacity index: it was no longer an indicator of effective-demand shortfalls to be corrected, but a sentinel for inflation control (Shaikh et al., 1999). High capacity utilization was redefined as a warning of upcoming price pressures, prompting monetary tightening to cool the labor market. Academic and policy debates increasingly evaluated capacity measures by their ability to predict inflation, subordinating employment targets to price-level stability (Klein et al., 1973). In these debates, Alan Greenspan argued that capacity measures must reflect systemic shortages rather than administrative averages, asserting that “if the numbers do not indicate that the economy is currently at capacity, then I suggest that the numbers are wrong, and that the correct concept of capacity must reflect the situation” (Klein et al., 1973, p. 758). This shift did not resolve the unobserved nature of the productive ceiling, but it established the technocratic foundation for modern output-gap governance.

2.2 From Capacity Utilization to Output-Gap Governance

2.6 During the late twentieth century, the rise of global trade and financialization reduced the policy relevance of national manufacturing surveys. Central banks and international financial institutions demanded portable, rule-based indicators to monitor economic activity across national borders (Jessop, 2002, 2003; Konings, 2007). This demand institutionalized the “output gap”—the difference between actual output and potential output. This transition replaced discretionary demand management with technocratic, rule-based governance models, such as inflation targeting, Taylor policy functions, and central bank independence (Taylor, 1993; Johnson, 2024). Under the New Consensus Macroeconomics, potential output was redefined as the non-inflationary limit

of economic activity, serving as an anchor to enforce wage discipline through NAIRU estimates (Saad-Filho, 2018). Yet this policy turn did not resolve the measurement problem: potential output remained a latent variable whose shape depended entirely on modeling assumptions. Like other technocratic indicators that mask political-economic conflicts behind fragile statistical choices (Herndon et al., 2014; Ash et al., 2017), potential output is treated by policymakers as a neutral, engineering-like benchmark that depoliticizes underlying distributional conflicts.

2.7 Potential output, however, is unobservable. Hence, its estimation is contingent on specific paradigms. For example, mainstream economics typically relies on statistical filters that smooth observed output trends (Hodrick and Prescott, 1997; Baxter and King, 1999; Beveridge and Nelson, 1981; Hamilton, 2018), aggregate production functions (Congressional Budget Office, 2004; Alich, 2015), or structural vector autoregressions (Blanchard and Quah, 1989). These aggregate methodologies function as a "black box," treating capacity formation as a friction-free engineering exercise. In reality, the conversion of accumulation demand into productive capacity is institutionally and distributionally mediated, shaped by conflict at the workplace and competitive pressures in the market. The identification of potential output depends on the intensity of labor and the organization of the workplace, where class struggle over work schedules, machine speed, and supervisor discipline mediates how intensively machines run. In the market, capital strikes and shifts in the reserve army of labor regulate whether this potential capacity is realized. When low unemployment strengthens labor's bargaining power, profit squeeze dynamics prompt capital to idle capacity rather than expand production.

2.8 The 2008 Global Financial Crisis exposed the structural limits of this technocratic framework. Post-crisis revisions routinely flipped the estimated sign of output gaps, and downward-revised potential-output estimates were used to justify fiscal austerity and compress public investment in the European periphery (Heimberger and Kapeller, 2017; Heimberger, 2020). While supranational research divisions, such as those of the IMF, documented severe hysteresis and revision volatility (Cerra and Saxena, 2017; Cerra et al., 2023), policy surveillance accelerated the use of high-frequency nowcasting and forward-guidance models to preserve the output-gap apparatus as adequate for real-time policymaking (Aastveit and Trovik, 2014; Foster, 2024). This revision-prone, closure-dependent policy framework remains entrenched, providing the institutional backdrop against which heterodox debates over capacity utilization must be understood.

2.3 Capacity-Utilization Debates in Political Economy

2.9 Within the debates in political economy, the "capacity utilization controversy" defines clear dividing lines between contested approaches under a central inquiry: is the long-run normal rate of utilization endogenous to demand, or is it anchored in cost structures and classical competition? The Neo-Kaleckian tradition treats the normal rate as endogenous, arguing that firms adjust their long-run operating rates in response to demand shocks, micro-level shift choices, or historical hysteresis (Lavoie, 1995; Nikiforos, 2013; Setterfield and David-Avritzer, 2020). In contrast, Sraffian-Keynesian economists argue that cost minimization and classical competition anchor the normal rate of utilization in the long run, with capital investment adjusting capacity to meet changes in effective demand (Ciccone, 1986; Kurz, 1986).

2.10 Both heterodox positions face conceptual and empirical challenges when identifying the productive ceiling. In the Sraffian supermultiplier framework, convergence to a normal rate requires a component of aggregate demand to remain permanently autonomous, a condition that is difficult to reconcile with the debt dynamics of stock-flow consistent models (Nikiforos, 2018). On the other hand, the Neo-Kaleckian claim of endogenous normal utilization depends on capacity utilization exhibiting non-stationarity (or path dependency) in the long run. This empirical premise has triggered a revival of the "capacity utilization controversy" in political economy. Critics like Gahn and González (2020) comment on Nikiforos's data, and Gahn and González (2022) present cross-country evidence, arguing that capacity utilization is stationary and mean-reverting, suggesting that actual utilization indeed converges to a structurally determined normal rate. However, these empirical tests remain highly contested, with their results depending entirely on the choice of data and unit-root testing specifications (Nikiforos, 2020).

2.11 By focusing primarily on manufacturing, both the FRB and AWW series fail to capture the structural shifts of modern, service-dominated economies (Haluska, 2023). Identifying the long-run dynamics of capacity formation therefore requires an alternative empirical approach that bypasses these survey-based and smoothing constraints entirely.

2.4 Shaikh's Structural Identification of Capacity Utilization

2.12 Rather than relying on survey measures or statistical filters that assume a symmetric business cycle and enforce mean reversion around a smooth potential trend, Shaikh (2016) shifts the identification of the capacity ceiling to classical profitability accounting. This methodology addresses a core conceptual limit of conventional filters: by smoothing out fluctuations, standard statistical routines treat expansions and contractions as mirror images, thereby obscuring the fundamental asymmetry of the business cycle—where slow, grinding booms are interrupted by rapid, violent contractions. This critique of symmetric filters aligns with Latin American neo-structuralist accounts by Ffrench-Davis (2010, 2012), who argue that deep contractions induce a persistent "recessive gap"—a demand-driven deficit that permanently scars productive capacity through hysteresis rather than mean-reverting. Both perspectives reject smoothing filters, showing that capacity utilization measurement must capture these asymmetrical paths. Shaikh's framework anchors capacity in the dynamics of capital accumulation and profitability. Using a Generalized Perpetual Inventory Method (GPIM) to construct the capital stock, he derives the potential-output path from the accounting identities linking actual profit rates to the maximum potential rate. Under this approach, capacity utilization is identified as the residual deviation of actual output from this long-run, profit-rate-determined capacity ceiling, preserving the historical and asymmetrical traces of accumulation and scrapping.

2.13 This accounting-based approach is methodologically significant because of its reproducibility and portability. By relying strictly on national accounts, alongside a theoretically grounded distinction of net capital stocks (capturing valuation dynamics) versus gross capital stocks (reflecting productive assets in operation), it provides a transparent and replicable framework for independent researchers to analyze accumulation dynamics without relying on country-specific survey designs or arbitrary statistical filters. However, treating capacity utilization as a deterministic residual assumes that output and capital share a stable, long-run cointegrating vector. If this underlying econometric relation is misspecified or unstable, the resulting capacity index will be highly sensitive to minor

specification choices. This chapter critically replicates Shaikh’s cointegration approach to stress-test its robustness. In doing so, it serves as a necessary stepping stone: by mapping the empirical boundaries and sensitivities of Shaikh’s baseline model, we lay the groundwork to overcome its limitations in subsequent chapters.

3 Conceptual Framework

3.1 The conceptual framework of capacity utilization rests on a fundamental accounting challenge: while aggregate output is directly observed, the productive ceiling remains a latent, unobservable variable. This measurement problem is deeply entangled with the critique of aggregate production functions. Rather than trying to identify physical laws of production or marginal productivities through neoclassical functions, we follow Shaikh’s methodology which estimates capacity utilization from the long-run relation between aggregate output and capital stock. Because this estimated path serves as the denominator for the capacity utilization index, the properties of the resulting series depend entirely on the stability of the output–capital coefficient driving the long-run relation between them. To analyze this capacity ceiling, we ground this relationship in an aggregate Leontief production structure with fixed labor-capital proportions, which is the standard toolkit of the political economy of growth and distribution modeling. This framework treats capacity utilization not as a constant, but as a variable output-capital ratio that deviates from its normal ceiling.¹

3.2 To map this conceptual framework, this section proceeds as follows. First, we analyze the capital-to-capacity relationship within an aggregate Leontief production structure. Second, we evaluate the algebraic identity trap highlighted by the accounting identity critique (Shaikh, 1974; ?), demonstrating the misspecification problem of omitting distribution when estimating value-based macroeconomic relations. Third, we relax the balanced-growth assumption to define accumulation regimes and analyze capital accumulation instability. Finally, we analyze the trend-stabilized closure, recalling Basu’s viability criteria and the double-misspecification problem in econometrics specification to show how exogenous trends mask the endogeneity of technical change and distribution.

3.1 Production Function and Productive Capacities

The empirical validity of a capacity-utilization series depends strictly on the stability of the parameter that generates the fitted capacity path. While Shaikh (2016) avoids direct physical proxies, his empirical output-capital relation can be grounded in a standard classical Leontief production function:

$$Y_t = \min\{A_t L_t, \mu_t R_t^n K_t\}, \quad (2)$$

3.3 where Y_t is aggregate output, L_t is employment, K_t is the capital stock, A_t is labor productivity, R_t^n is the unobserved normal capacity-capital ratio (which is Shaikh’s original symbol for capital productivity at normal capacity utilization, Y_t^p/K_t), and μ_t is capacity utilization. In a capital-constrained regime where labor is not binding due to the reserve army of labor, this production function simplifies to $Y_t = \mu_t R_t^n K_t$. Taking natural logarithms yields:

¹Although we treat the capacity transformation elasticity, θ , as a technological parameter in this chapter, subsequent chapters will endogenize θ to show how distributive struggles and changes in the wage share shape the long-run capacity ceiling.

$$\ln Y_t = \ln K_t + \ln R_t^n + \ln \mu_t. \quad (3)$$

This aggregate Leontief specification hides a concrete workplace-level transmission mechanism. At the shop-floor level, the conversion of the capital stock K_t into potential capacity ($Y_t^p = R_t^n K_t$) is not a passive technical constant. Rather, it is determined by the speed of machinery, the organization of shifts, and the physical intensity of labor. This conversion rate, captured in the normal capacity-capital ratio R_t^n , changes over time as capitalists introduce labor-saving technical modifications to increase labor discipline. Once potential capacity is established, the production of actual output Y_t is mediated by capacity utilization μ_t , which reflects the pace of market demand. Class struggle over the length and intensity of the workday directly moderates these conversions, determining both the maximum potential output capitalists can extract from a given capital stock and the actual rate of utilization they can sustain.

3.4 Isolating the long-run capacity path requires modeling the unobserved normal capacity-capital ratio ($\ln R_t^n$). [Shaikh \(2016\)](#) proposes a linear function of a deterministic time trend and the observed capital stock:

$$\ln R_t^n = bt + c \ln K_t. \quad (4)$$

3.5 Substituting this definition into the Leontief-derived log relation yields the estimable single-equation level relation:

$$y_t = a + bt + d k_t + \varepsilon_t, \quad d \equiv 1 + c, \quad (5)$$

3.6 In this regression, y_t and k_t represent the natural logarithms of unscaled output and capital, while d represents the long-run output-capital elasticity, which this chapter reinterprets as the unbalanced growth transformation elasticity θ . The constant a absorbs initial conditions and fixed measurement discrepancies in the levels of output and capital. The linear trend component (bt) is conventionally interpreted as a measure of autonomous technical change. Whereas Shaikh focuses on the error-correction term, interpreting it as the capacity utilization rate, this chapter focuses instead on the underlying long-run behavior driven by the relation between capital accumulation and productive capacity creation ($g_{Y^p} = \theta g_K$). Consequently, the deterministic trend included in a cointegrating level relation might obscure an omitted variable bias imposed by this theoretical misspecification.

3.7 The fitted path of productive capacity follows directly from the recovered long-run components:

$$\hat{y}_t^p = \hat{a} + \hat{b}t + \hat{d} k_t. \quad (6)$$

3.8 Capacity utilization is then constructed as the deviation of realized output from this long-run capacity path:

$$\ln \hat{\mu}_t = y_t - \hat{y}_t^p, \quad \hat{\mu}_t = \frac{Y_t}{\hat{Y}_t^p}. \quad (7)$$

3.9 This estimation sequence establishes a direct functional dependency, where the constructed utilization series inherits its level, timing, and variance from the estimated parameter $\hat{a}$. Consequently, different single-equation specifications yield divergent $\hat{a}$ estimates, producing contradictory

trajectories for both capacity and utilization. Because the model forces a dynamic accumulation process into a rigid accounting identity, the empirical sensitivity of this parameter raises an immediate identification problem. Structural measurement therefore rests on evaluating whether this long-run multiplier captures a genuine behavioral law of production, or whether it operates merely as an algebraic placeholder that absorbs omitted historical variation.

3.2 Laws of Algebra and Laws of Production

3.10 The estimated long-run coefficient d operates as the transformation elasticity ($\theta \equiv \partial \ln Y^p / \partial \ln K$) anchoring the profit-rate decomposition of [Shaikh \(2016\)](#). Because the normal rate of profit relies directly on the constructed capacity–capital ratio, the structural integrity of the entire empirical operationalization of his theoretical contributions rests on this specific parameter. While the aggregate accounting identity deterministically organizes these profit magnitudes, it leaves the underlying behavioral mechanism translating capital accumulation into productive capacity undefined. Following the analytical standard of [Shaikh \(1974\)](#), an accounting identity that leaves its core behavioral mechanism undefined mechanically mimics a structural production law. As previously stated, this entails a misspecification risk: the inclusion of a trend variable might partially absorb historical and institutional variation overlooked in Shaikh’s formulation.

3.11 Evaluating Shaikh’s capacity regression reveals that it is vulnerable to the very critique he leveled against Solow’s aggregate production function. In his seminal critique, [Shaikh \(1974\)](#) showed that the high fit of aggregate production functions is a mathematical byproduct of the underlying value-added accounting identity ($p_t Y_t \equiv W_t + P_t = w_t L_t + r_t K_t$) under a constant wage share, rather than proof of physical substitution laws. As elaborated by [?](#), aggregate regressions using constant-price monetary data do not identify physical production laws, but instead approximate this value identity. Applying this critique to Shaikh’s own capacity regression reveals a similar limitation: since the output–capital ratio is linked by identity to the rate of profit (r_t) and the profit share (π_t) via $Y_t/K_t = r_t/\pi_t$, taking logarithms yields $y_t - k_t = \ln r_t - \ln \pi_t$. When we estimate the level relation over a period of shifting profit shares and class struggle, the deterministic trend bt does not necessarily identify an autonomous technological change component, but instead reflects distributive variation. The long-run coefficient d must adjust to satisfy the accounting identity, mimicking a stable physical relation while masking changes in the rate of exploitation.

3.12 To ground this value identity trap in post-war US history using the original data tables by [Shaikh \(2016\)](#), we observe that while real gross value added grew at an average annual rate of 3.3% and real gross capital stock grew at 4.4%, the output-to-capital ratio exhibited long-run stability but fluctuated significantly around a historical average of 0.53. More importantly, this variation is closely tied to shifts in the rate of exploitation ($e_t = \text{NOScorp}/\text{ECcorp}$), which acts as the empirical proxy for the rate of surplus value. During the Post-War Golden Age (1947–1973), the rate of exploitation was high and stable, averaging 0.319 (corresponding to a net profit share of 21.78%). In the subsequent stagflation transition (1974–1982), a structural profit squeeze caused the exploitation rate to collapse to a historical low of 0.256 (profit share of 18.54%). Finally, during the Neoliberal Era (1983–2011), the containment of real wage growth restored the exploitation rate to an average of 0.286 (profit share of 20.13%). Because the single-equation ARDL model excludes these distributional shifts, the estimated long-run technical coefficient is contaminated by this omitted historical variation, appearing stable only because it mathematically averages across

these distinct regimes of class struggle. This confirms that the output–capital relationship is not a static engineering baseline, but is dynamically conditioned by the rate of exploitation.

3.13 The capital stock series K_t , reconstructed via Shaikh’s Generalized Perpetual Inventory Method (GPIM) to correct for infinite service-life biases, converges to a 99.6% average ratio with the BEA current-cost measure (1947–2005); the remaining measurement discrepancy is therefore largely transitory and stationary. This discrepancy is not mere statistical noise; it reflects the material imprint of workplace dynamics—deferred maintenance, extended machine running times, and labor speed-ups—that allow physical capital to outlast its book-value depreciation schedule. Because this error is stationary, standard cointegration theory dictates it does not induce systematic asymptotic bias in the estimated cointegrating vector, though any finite-sample distortion cannot be signed *a priori*. We interpret our empirical estimates of θ conservatively, recognizing that the true structural conversion of capital into active productive capacity may be either somewhat more or less responsive than the baseline data suggests.

3.3 Balanced vs. Unbalanced Growth: Accumulation Regimes and Capital Accumulation Instability

3.14 Marx’s schemas of reproduction provide the analytical benchmark for proportional macroeconomic growth (Okishio, 2022). Smooth capitalist reproduction requires strict material proportions: commodities must sell, surplus value must be realized, and the material composition of output must match the input requirements of continuous capital accumulation (Basu, 2022). Because real capitalist economies routinely violate these proportions, smooth reproduction remains an analytical abstraction rather than a historical description. To isolate the supply-side relation between capital and capacity, the long-run classical closure ($\mu = 1$) holds demand instability constant (Foley, 1985). Alternatively, canonical post-Keynesian models enforce a balanced-growth baseline by assuming a unitary transformation elasticity ($\theta = 1$) in the long run. Combined with demand fluctuations, this unitary assumption generates the characteristic properties of the Harrodian knife-edge.

3.15 Introducing an unbalanced growth closure ($\theta \neq 1$) achieves long-run identification by treating proportional accumulation as suggested by Okishio (2022), as a restrictive special case of equilibrium accumulation trajectory. Relaxing the unitary assumption shifts the analytical focus away from the classical closure ($\mu = 1$). Under a non-unitary parameter, systematic instability is an inherent feature of capacity formation, distinct from temporary deviations driven by demand. The transformation elasticity (θ) governs the long-run development of productive capacity independently of short-run demand stabilization. In this setup, θ acts as a structural parameter, where the interest of this critical replication lies. Its endogenization to distributive conflict is beyond the scope of this chapter, but will be tackled in the next chapter of this dissertation.

3.16 Defining the transformation elasticity as $\theta \equiv \partial \ln Y^p / \partial \ln K$, the analytical properties of the accumulation system depend on whether this parameter deviates from unity. Table 1 defines three macroeconomic regimes bounded by these conditions.

Table 1: Accumulation Regimes and the Productive-Capacity Relation

Regime	Parameter Condition	Long-Run Relation ($\hat{Y}^p, \hat{K}$)	System Tendency
Balanced Growth	$\theta = 1.0$	Capital and capacity expand proportionally	Knife-edge baseline
Overaccumulation	$\theta < 1.0$	Capital expands faster than productive capacity	Accumulation decelerates toward stagnation
Excess Capacity	$\theta > 1.0$	Productive capacity expands faster than capital	Accumulation accelerates without supply-side limits

Note: This table defines the theoretical regimes governing the system.

3.17 We frame this regime classification under a single-sector macroeconomic bounding; while this simplifies the analysis of stagnation tendencies driven by overaccumulation ($\theta < 1$), it represents a key analytical boundary. The debates around Marx’s reproduction schemas on disproportionality emphasize the material and value balances between Department I (means of production) and Department II (means of consumption). By abstracting from these sectoral flows, our single-sector framework reframes the concept of disproportionality to the structural decoupling of capital accumulation and capacity expansion in terms of unbalanced growth: between capital stock and productive capacities, under the assumption that the only source of demand that builds productive capacities is accumulation demand. Unbalanced growth is represented by a non-unitary transformation elasticity ($\theta \neq 1$). This serves as a simplified baseline to work under a single-sector macroeconomic framework for our empirical replication and subsequent chapters of the dissertation, while acknowledging that sectoral imbalances are abstracted from our analysis.

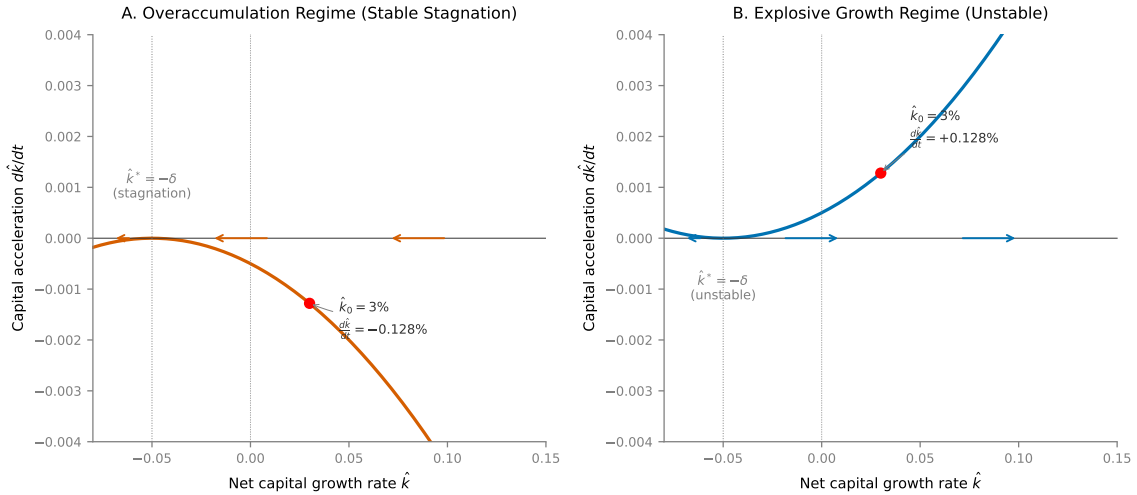
3.18 These regimes represent the analytical limits of the accumulation process. Under an unbalanced growth closure without autonomous technical change, the dynamic path of the net capital growth rate ($\hat{k} \equiv \dot{K}/K - \delta$) is governed by an ordinary differential equation where the sign of $d\hat{k}/dt$ depends on the magnitude of $(\theta - 1)$ (the formal mathematical derivation and stability proofs are provided in Appendix A). Net accumulation decelerates under $\theta < 1$, accelerates under $\theta > 1$, and remains stationary under $\theta = 1$. This instability reflects the accounting laws of capital accumulation under a non-unitary transformation elasticity, distinct from Harrodian demand-driven dynamics.

3.19 Formally, $d\hat{k}/dt$ is a capital growth acceleration term, representing the rate of change of the net capital accumulation rate over time. Because a perpetual acceleration or deceleration of capital growth is economically impossible over long horizons, this term must remain bounded near zero over the long run. In historical capitalist economies, this bounding is enforced by distinct institutional and macroeconomic limits depending on the accumulation regime. Under an excess capacity regime ($\theta > 1$), a one percent rise in the capital stock generates a proportionally larger expansion in productive capacity. Theoretically, this could sustain a stable, high-growth convergence with elevated capacity utilization only if continuously offset by an exogenous source of autonomous demand (such as government expenditures or exports). Absent such demand, the system faces chronic excess capacity and realization crises, which eventually bound the acceleration through debt deflation, capacity scrapping, and investment collapses. Conversely, under an overaccumulation regime ($\theta < 1$), capital accumulation outpaces capacity formation, causing the system to hit hard

supply-side bottlenecks. This triggers rate-limiting factors such as labor cost inflation (wage barriers), central bank interest rate hikes (monetary policy shifts), and financial leverage limits, which choke off the expansion and force the acceleration term back toward zero.

3.20 To illustrate this dynamic instability qualitatively using the original replication data structures of [Shaikh \(2016\)](#), consider a stylized example under both overaccumulation and explosive growth regimes. Let the baseline net capital growth rate be $\hat{k}_0 = 3\%$ (0.03) and the depreciation rate be $\delta = 5\%$ (0.05), implying a gross growth rate of capital of 8%. If we assume a unitary transformation elasticity ($\theta = 1$) under a balanced-growth baseline, the capacity ceiling expands proportionally at 8% and the net accumulation rate remains stationary ($d\hat{k}/dt = 0$). However, if the transformation elasticity is non-unitary, say $\theta = 0.8$, the system experiences overaccumulation. Here, a 1% growth in capital relative to capacity formation entails a 0.2% capacity mismatch ($\theta - 1 = -0.2$). This mismatch induces a persistent downward drag on capital accumulation (normal capital productivity decays at $\hat{R}^n = -1.6\%$), decelerating the net accumulation rate at a rate of $d\hat{k}/dt = -0.128\%$ per year toward stagnation. Conversely, if the transformation elasticity is $\theta = 1.2$, capacity expands at 9.6%, causing normal capital productivity to rise ($\hat{R}^n = +1.6\%$) and the net accumulation rate to accelerate at $d\hat{k}/dt = +0.128\%$ per year, driving the system into an unbounded, explosive trajectory (as detailed in Appendix A). This dynamic instability is illustrated in Figure 1, which plots the phase diagram for both regimes.

Figure 1: Capital accumulation dynamics under unbalanced growth



Note: Panel A plots the overaccumulation regime ($\theta = 0.8 < 1.0$) where capital acceleration is negative ($d\hat{k}/dt < 0$ for $\hat{k} > -\delta$), pushing the net accumulation rate toward the stable stagnation equilibrium at $\hat{k}^* = -\delta$. Panel B plots the explosive growth regime ($\theta = 1.2 > 1.0$) where capital acceleration is positive ($d\hat{k}/dt > 0$ for $\hat{k} > -\delta$), causing the accumulation rate to diverge. Both panels show the initial condition $\hat{k}_0 = 3\%$ where $d\hat{k}/dt = \mp 0.128\%$, consistent with the numerical example in §3.3.

3.21 While effective demand and distributive conflict are crucial to the broader macroeconomic picture, we abstract from them in this initial formulation to isolate the supply-side mechanics of accumulation. Institutional and macroeconomic forces certainly influence the speed and direction of capital accumulation, but they do not generate the underlying dynamic itself. The differential

equations in Appendix A demonstrate how the system behaves under different accumulation regimes. However, the empirical single-equation model we will estimate restricts the transformation elasticity (θ) to a constant value over time. The next step is to test whether this assumption of a time-invariant θ actually holds in the historical data, thereby revealing the limits of this fixed-parameter approach.

3.4 Trend Stabilization, Viability Condition, and the Double-Misspecification Bias

3.22 We have previously argued that, econometrically, the inclusion of a time trend might absorb variation arising from distributive conflict and distinct historical configurations. Hence, we analyze the inclusion of a time trend under an unbalanced growth closure ($g_{Y^p} = \theta g_K$) and its consequences for dynamic stability. As established in the phase dynamics, a system experiencing overaccumulation ($\theta < 1$) forces the net accumulation rate to decay to zero. Using the same accounting identities that drive dynamic stability, we can show that introducing a deterministic level trend (b) under an unbalanced growth closure ($g_{Y^p} = \theta g_K + b$) stabilizes the system.

3.23 Substituting this extended closure into the acceleration identity transforms the system into a quadratic ordinary differential equation:

$$\frac{d\hat{k}}{dt} = (\hat{k} + \delta) \left[(\theta - 1)\hat{k} + b \right]. \quad (8)$$

3.24 This specification shifts the equilibrium structure of the accumulation process. The stagnation limit ($\hat{k}^* = 0$) is replaced by a new interior root:

$$\hat{k}^* = \frac{b}{1 - \theta}. \quad (9)$$

3.25 Under overaccumulation ($\theta < 1$) with positive autonomous technical progress ($b > 0$), this interior equilibrium is locally stable. The depreciation rate (δ) anchors the negative root, while the ratio $b/(1 - \theta)$ scales autonomous progress against the structural imbalance, preventing stagnation. Conversely, in the excess capacity regime ($\theta > 1$), the deterministic trend b fails to provide a stabilizing anchor. Because $(1 - \theta)$ is negative, the interior root $\hat{k}^* = b/(1 - \theta)$ becomes negative, meaning no economically meaningful positive steady-state accumulation rate exists. Instead of stabilizing the system, the trend b merely compounds the dynamic divergence, exacerbating the explosive trajectory where productive capacity continuously outpaces capital accumulation and reinforcing the chronic excess capacity that ultimately requires exogenous autonomous demand to resolve.

3.26 The analysis of technical change by Basu (2022) demonstrates why treating the trend-stabilized steady state as a smooth, exogenously determined path is fundamentally problematic. Basu establishes that under capitalism, technical change is not a neutral, exogenous drift but is governed by the strict filter of *capitalist viability*: producers only adopt new techniques if they reduce costs at prevailing prices and real wages. Crucially, the trajectory and specific type of technical change—whether it is capital-using, labor-saving (CU-LS), or capital-saving, labor-saving (CS-LS)—are not predetermined. Instead, they are endogenously shaped by the balance of class struggle over the real wage relative to productivity growth. Therefore, the deterministic trend

coefficient (b) in the capacity-capital ratio does not represent autonomous technological progress. Rather, it is the econometric residue of the feedback loop between distributional conflict and the choice of technique. In other words, imposing a time trend masks how class struggle actively and endogenously reshapes the ceiling of productive capacities.

3.27 The statistical validity of the capacity utilization series relies on establishing a stable cointegrating relation between output, capital, and the deterministic trend. Omitting distributive conflict while including the exogenous trend introduces a double misspecification. As Basu (2020) demonstrates, when an econometric model excludes a relevant variable—in this case, distributive conflict and its influence on the choice of technique (Kurz, 1986)—while including an irrelevant proxy like a deterministic time trend, the estimator for the capital coefficient suffers from compound bias. Because the trend mathematically absorbs the unobserved shifts in distribution, this indirect bias channel compromises the recovered elasticity parameter θ , making it a passive statistical fit rather than a stable production relation.

3.28 Estimating this relation over Shaikh’s sample imposes rigid temporal constraints. This restriction forces both the elasticity (θ) and the trend (b) to act as time-invariant constants across shifting institutional configurations. The classical long-run closure ($\mu = 1$) isolates theoretical capital-to-capacity relations (Foley, 1985) or smooth reproduction paths (Okishio, 2022; Basu, 2022). Because real capitalist economies routinely violate these strict proportionality conditions, imposing time-invariance forces the recovered parameters to mathematically average across structurally heterogeneous historical periods.

3.29 Looking at the actual post-war historical trajectory of the US corporate sector (1947–2011) using the original data tables by Shaikh (2016), these theoretical regimes correspond to distinct accumulation phases. During the Post-War Golden Age (1947–1973), the US corporate sector experienced rapid capital accumulation alongside strong output growth, with the real gross capital stock expanding at an average annual rate of 5.4% and real gross value added growing at 4.4%. This period represents a relatively balanced accumulation path with a high, stable output–capital ratio. During the transition of the stagflation crisis (1974–1982), the growth of the real capital stock remained relatively high at 4.8% (driven by energy-related investments and structural restructuring), while real output growth slowed significantly to 2.7%, causing the output–capital ratio to collapse. Subsequently, in the neoliberal era (1983–2011), the rate of capital accumulation fell to an average of 3.3% annually, while real output grew at only 2.7%, leaving the output–capital ratio volatile and structurally depressed. This empirical shift in how capital accumulation translates to productive capacity strongly suggests a change in the underlying transformation elasticity θ across these eras. Imposing a time-invariant parameter over this entire 64-year span forces the econometric model to mathematically average across these structurally distinct historical periods, actively masking the very crises and regime shifts that define capitalist development.

3.30 This parametric averaging also bears directly on the classical debate concerning the long-run endogeneity of capital accumulation to aggregate demand. In the Neo-Kaleckian framework, capacity utilization is endogenous to demand in the long run, and investment adjusts to minimize utilization gaps. Conversely, the Sraffian position (e.g., the Supermultiplier model) treats the choice of technique and the capital accumulation path as weakly exogenous to demand-led capacity disequilibrium,

adjusting instead to cost-minimizing criteria at a given real wage. Our conceptualization of unbalanced growth ($\theta \neq 1.0$) introduces a more nuanced, counteracting dynamic to this debate. Under overaccumulation ($\theta < 1.0$), capital accumulation is structurally unstable, meaning that the capacity ceiling grows slower than capital stock. While this decoupling implies that investment is weakly exogenous to short-run capacity utilization errors, the system-level instability prevents simple long-run demand adaptation. Crucially, if the weak exogeneity of investment and the secular instability of overaccumulation act as counteracting forces, they may cancel each other out. This counteracting dynamic represents a plausible interpretation that might obscure the resolution of the Sraffian and Neo-Kaleckian debates themselves—an analytical blind spot explained by both frameworks operating under the restrictive assumption of a balanced-growth path with a fixed output–capital ratio.

3.31 The next section 4 implements an analytical stress test across three distinct stages to identify the conditions under which Shaikh’s single-equation ARDL estimation fails to identify cointegration. Stage S0 establishes a baseline of reproducibility using the original data tables by Shaikh (2016) and lag lengths. Stage S1 implements an admissibility test by estimating a grid of 500 ARDL specifications, mapping parameter sensitivity to researcher choices—such as lag structures, information-criterion penalties, and historical dummy variables. Stage S2 estimates the long-run relationship using a Vector Error Correction Model (VECM) to determine whether joint system stability holds in a bivariate setting, or if it requires the logged rate of exploitation ($e_t = \ln[\pi_t/(1 - \pi_t)]$) to establish a stable cointegrating relationship.

4 Critical Replication Study on Shaikh’s Approach to Cointegration

4.1 Econometric Framework and Staged Design

4.1 Shaikh’s long-run output–capital coefficient does not identify a stable, time-invariant capacity benchmark; it mechanically averages over historically distinct distributive regimes. As established in Section 3, this coefficient functions as a transformation elasticity (θ) linking capital accumulation to capacity formation under conditions of unbalanced growth. This section tests whether a single, full-sample estimation can capture this relationship, or whether the bivariate relation is fundamentally misspecified without explicit distributional conditioning. By testing the cointegrating relation directly rather than relying on a pre-fabricated utilization index, we subject the underlying classical capacity theory to empirical testing.

4.2 We execute this critical investigation across three sequential stages of increasing econometric generality. In Stage S0, we reconstruct the closest empirical counterpart to Shaikh’s baseline single-equation model to verify its basic numerical reproducibility. Stage S1 lifts the baseline’s rigid specification choices by estimating a combinatoric grid of 500 Autoregressive Distributed Lag (ARDL) models, mapping how the estimated transformation elasticity ($\hat{\theta}$) shifts under alternative lag structures, trend configurations, and outlier controls. Finally, Stage S2 transitions the analysis from single-equation models to system-level estimations, testing whether the output–capital relation survives as a stable Vector Error Correction Model (VECM) and checking whether it requires the rate of exploitation (e_t) to establish system-level cointegration.

4.3 Our findings show that while Shaikh’s baseline coefficient is numerically reproducible, it is neither structurally unique nor stable in isolation. In the single-equation grid search, the estimated elasticity ranges from 0.65 to 0.92 depending on how we penalize model complexity, indicating that the baseline point estimate is highly sensitive to researcher choices. More importantly, the standalone output–capital relation fails system-level cointegration tests, proving that a stable long-run relationship cannot be identified in a bivariate setting. The output–capital relation achieves system stability only within a trivariate framework that includes the rate of exploitation, showing that the long-run conversion of capital to capacity is structurally conditioned by the historical distribution of income between wages and profits.

4.4 We present these results sequentially across seven subsections. First, Subsection 4.1 outlines the mathematical framework and staged design. Subsection 4.2 details the data construction, measurement conventions, and stylized facts. Subsection 4.3 defines the admissibility filters and diagnostic screens. Subsections 4.4, 4.5, and 4.6 present the empirical results for Stages S0, S1, and S2, respectively. Finally, Subsection 4.7 provides a cross-stage econometric synthesis.

4.5 We formalize this progressive, multi-stage testing sequence in Table 2, which outlines the design object, specification space, and diagnostic criteria for each stage. By moving from a single-point reconstruction (S0) to a combinatorial specification grid (S1), we systematically map how parameter estimates react to changes in the model’s complexity. The transition to joint system estimation (S2) then tests whether these single-equation relationships survive as self-sufficient economic attractors once we allow for bidirectional feedbacks. This staged architecture allows us to separate the statistical fit of the model from the historical and institutional validity of its derived capacity path.

Table 2: Stage Design and Admissibility Strategy

Stage	Design Object	Specification Space	Admissibility and Organization	Interpretive Implication
S0	Single-point reconstruction of baseline specification	ARDL(2,4), historical impulse dummies, alternative deterministic closures	F-bounds primary screen; t-bounds diagnostic where defined; utilization path proximity	Tests whether a deterministic capacity path can be closely reconstructed
S1	ARDL specification grid $\mathcal{G}_{S1}$	Lag orders (p, q) , five PSS cases, four nested dummy structures	F-admissible set $\mathcal{A}_{S1}^F$; F+t confirmed candidates $\mathcal{C}_{S1}$; corrected envelope E_{S1} ; IC-specific neighborhoods $\mathcal{F}_j^{(0.20)}$	Tests whether the output–capital coefficient is structurally unique or varies with IC penalty logic
S2 Bivariate	Joint $(\ln Y_t, \ln K_t)'$ VECM	Lag order, deterministic branch, dummy structure, $r = 1$	Triple gate $\mathcal{A}_{S2}$: convergence, Johansen rank, companion stability	Tests whether the bilateral output–capital relation is self-sufficient at the system level
S2 Trivariate	Joint $(\ln Y_t, \ln K_t, \ln e_t)'$ VECM	Same as bivariate, with logged rate of exploitation included; $r = 1$ or $r = 2$	Same triple gate; Ω_{20} organizes admissible survivors by log-likelihood loss	Tests whether the structural relation survives exclusively when distribution enters explicitly

Note: This table outlines the progressive multi-stage replication architecture. The sequence advances from baseline single-equation reconstruction (S0) through parameter sensitivity grids (S1) to joint system-level survival testing (S2). Source: Author’s elaboration.

4.6 We write the output–capital relation as an Autoregressive Distributed Lag (ARDL) model to identify the long-run transformation elasticity (θ). To evaluate whether output (y_t) and capital (k_t) share a stable long-run equilibrium, we use the bounds-testing procedure developed by [Pesaran et al. \(2001\)](#). Cointegration means that although output and capital drift over time as nonstationary series, they are tied together by a structural attraction force that prevents them from drifting apart indefinitely. This framework detects this cointegrating relationship regardless of whether the individual series are integrated of order zero ($I(0)$) or one ($I(1)$), eliminating the need for fragile unit-root pre-testing.

4.7 We estimate this relationship using an unrestricted error correction model (UECM) derived from a general ARDL(p, q) specification:

$$\Delta y_t = c_0 + c_1 t + \pi_y y_{t-1} + \pi_k k_{t-1} + \sum_{i=1}^{p-1} \psi_{y,i} \Delta y_{t-i} + \sum_{j=0}^{q-1} \psi_{k,j} \Delta k_{t-j} + \sum_{h=1}^H \delta_h D_{h,t} + \varepsilon_t. \quad (10)$$

4.8 This unrestricted error-correction representation separates the short-run adjustment dynamics from the long-run level relationship. In equation (10), c_0 is the constant intercept, $c_1 t$ is a deterministic linear time trend, and ε_t is an independent and identically distributed error term. The differenced terms Δy_{t-i} and Δk_{t-j} capture short-run memory and cyclical frictions, where p and q represent the lag lengths of output and capital. The lagged level terms y_{t-1} and k_{t-1} contain the long-run cointegrating relationship. The historical impulse dummies $D_{h,t}$ act as short-run shock absorbers, preventing outlier events (like the 1974 oil shock) from distorting the underlying long-run relationship.

4.9 The statistical distribution of the bounds test depends on how we restrict the constant and linear trend in the model. [Pesaran et al. \(2001\)](#) outline five deterministic configurations (Cases I through V) that determine how the constant c_0 and time trend t enter the equation. These cases range from Case I (no constant, no trend) to Case V (unrestricted constant and unrestricted trend), including intermediate cases where constants or trends are restricted to the long-run cointegrating space. Since these deterministic components alter the asymptotic behavior of the nonstationary variables, each case requires a different set of critical value bounds. We treat these five configurations as specification choices that alter the statistical boundaries of our cointegration test.

4.10 We use a joint Wald F -bounds test to evaluate the null hypothesis that no long-run relationship exists between output and capital. The null hypothesis (H_0^F) asserts that the coefficients on the lagged level terms are jointly zero, meaning there is no cointegration:

$$H_0^F : \pi_y = \pi_k = 0, \quad H_a^F : (\pi_y, \pi_k) \neq (0, 0). \quad (11)$$

4.11 Under this hypothesis, π_y and π_k are the coefficients on the lagged levels of output and capital. If the null is true, the regression is spurious because the residuals remain nonstationary ($I(1)$), meaning we find no evidence of cointegration, rendering any derived capacity path econometrically invalid. To evaluate the Wald statistic, we use the critical bounds generated by [Pesaran et al. \(2001\)](#). These bounds establish two thresholds: a lower bound assuming all variables are $I(0)$, and an upper bound assuming all variables are $I(1)$. We reject the null of no cointegration only when

the computed F -statistic exceeds the upper bound, confirming that the output–capital relation has a stationary residual.

4.12 We apply the t -bounds test as a secondary confirmation screen to ensure that the cointegrating relationship is stable. The Wald F -test can sometimes find pseudo-cointegration if only one of the lagged level variables is significant. The t -bounds test evaluates the null hypothesis that the coefficient on the lagged dependent variable (π_y) is zero:

$$H_0^t : \pi_y = 0, \quad H_a^t : \pi_y < 0. \quad (12)$$

4.13 We test this against the one-sided alternative hypothesis ($\pi_y < 0$), which requires the error-correction coefficient to be negative. A negative speed-of-adjustment coefficient acts as a restoring force: when actual output drifts away from its long-run capacity path, the system adjusts by pulling output back toward the equilibrium. If π_y is positive or zero, the system is explosive or has no memory. Given our relatively short sample period ($T \approx 65$), we use finite-sample critical bounds from [Natsiopoulou and Tzeremes \(2022\)](#) rather than asymptotic tables to prevent over-stating cointegration significance.

4.14 To recover the long-run transformation elasticity (θ), we must distinguish the underlying levels ARDL model from its error-correction representation. We write the general levels ARDL(p, q) specification as:

$$y_t = a_0 + a_1 t + \sum_{i=1}^p \alpha_i y_{t-i} + \sum_{j=0}^q \beta_j k_{t-j} + \sum_{h=1}^H \delta_h D_{h,t} + \varepsilon_t. \quad (13)$$

From this levels model, the long-run multiplier (d) is recovered as the sum of the capital stock coefficients normalized by the output lag coefficients:

$$\hat{d} = \frac{\sum_{j=0}^q \hat{\beta}_j}{1 - \sum_{i=1}^p \hat{\alpha}_i}. \quad (14)$$

This levels ratio represents the steady-state multiplier where the dynamic adjustment processes have fully settled. In his baseline estimation, Shaikh treats this parameter as a constant technical coefficient mapping capital directly to potential capacity. Under our conceptual framework, however, we interpret this long-run coefficient as a capacity transformation elasticity, θ , which is historically and distributionally conditioned.²

4.15 We map this levels specification directly to the unrestricted error correction model (UECM) in Equation (10) by reparameterizing the level variables around their first-difference representations. In the UECM, the speed-of-adjustment coefficient π_y and the level capital coefficient π_k relate to the levels parameters by $\pi_y = -(1 - \sum_{i=1}^p \alpha_i)$ and $\pi_k = \sum_{j=0}^q \beta_j$. Thus, the long-run multiplier is

²Shaikh’s original exposition ([Shaikh, 2016](#)) seldom discusses the parameter d as an elasticity, treating it instead as a fixed accounting coefficient. We interpret this long-run multiplier d as the empirical proxy for the capacity transformation elasticity θ under conditions of unbalanced growth. For clarity, we retain the notation d when replicating Shaikh’s ARDL specifications directly, and transition to θ when developing our conceptual and system VECM extensions.

recovered from the conditional UECM parameters as the ratio:

$$\hat{d} = -\frac{\pi_k}{\pi_y}. \quad (15)$$

This UECM ratio is mathematically identical to the levels ratio in Equation (14). A common pitfall in error-correction estimations is to calculate this long-run multiplier using the short-run differenced parameters from the UECM rather than the levels parameters. We ensure that only the levels coefficients enter the long-run relation to avoid such bias, correcting the notation to represent the true long-run cointegrating levels relation.

4.16 In Shaikh’s procedure, this long-run multiplier anchors the fitted capacity path. Capacity utilization is then the residual deviation from the long-run cointegration vector: $\log \hat{\mu}_t = y_t - \hat{y}_t^p$. This design makes utilization dependent on the estimated coefficient. Shaikh treats the coefficient as a fitted accounting multiplier, but he does not theorize its structural role. We re-interpret the coefficient as a transformation elasticity $\hat{\theta}$ of productive capacities with respect to capital accumulation, subject to the specification grid, deterministic treatment, and admissibility gates. We evaluate this elasticity across three stages: we reconstruct the baseline result (Stage S0), open the single-equation specification space (Stage S1), and test whether the relation survives in a joint system (Stage S2). Subsection 4.2 details the data construction and measurement conventions required to execute these steps. An econometric limitation of this single-equation setup is that if the capital stock (k_t) is measured with error due to the approximations of the Perpetual Inventory Method, the estimated elasticity $\hat{\theta}$ will be biased downward. Measurement error occurs because physical capacity changes are mediated by workplace organization, labor intensity, and speedups, which standard NIPA capital stock metrics cannot capture. The replication evaluates this baseline point model while explicitly testing this parameter’s stability across a broader specification grid.

4.17 In Stage S0, we test the reproducibility of Shaikh’s baseline results to establish a benchmark for our replication. We target his published single-equation model, fix the lag structure at the reported ARDL(2,4) configuration, and estimate the parameters using the closest recoverable corporate-sector data. We identify the variables directly from his published data tables, resolving documentation discrepancies in the original text. We estimate these baseline parameters across alternative deterministic configurations and historical impulse controls to reconstruct his point estimates, rather than to search for alternative specifications.

4.18 This procedure estimates the long-run relationship, constructs the potential capacity path, and calculates capacity utilization as the residual deviation of actual output from that path. We use cointegration bounds tests to distinguish genuine long-run relations from spurious correlations generated by autoregressive filtering. Here, we encounter a technical limitation of the ARDL framework: while the F -bounds test applies to all five deterministic cases, the t -bounds test is only defined for Cases I, III, and V. Specifications under Case II or Case IV (which restrict the constants or trends to the cointegration space) cannot be screened using the t -bounds gate.

4.1.1 Stage 1 (S1): Admissible-Specification Stress Test

4.19 In Stage S1, we expand the single-equation model into a combinatoric grid of 500 different specifications. By varying the lag lengths, deterministic trends, and outlier controls, we evaluate how

researcher choices affect the cointegration test results and the estimated elasticity. This approach shifts our focus from a single preferred model to the entire space of possible specifications, showing how sensitive the capacity path is to modeling assumptions.

4.20 We allow the output and capital lags to vary freely from 1 to 5 ($p, q \in \{1, \dots, 5\}$). We estimate each lag combination across all five deterministic cases, combined with four alternative blocks of historical dummy variables (s_0 through s_3). These dummies control for major post-war shocks: the 1956 Eisenhower recession, the 1974 oil crisis, and the 1980 Volcker monetary contraction. Permuting these options yields 500 unique models. Formally, the specification space is:

$$\mathcal{G}_{S1} = \{(p, q, c, s)\}, \quad (16)$$

4.21 In this grid $\mathcal{G}_{S1}$, each element (p, q, c, s) represents a single model. The indices p and q set the lag orders, c determines the PSS case configuration, and s indicates which outlier controls are active.

4.22 Admissibility screens filter out models that fail cointegration tests, but they do not identify a single correct parameter. The remaining models reveal a disciplined non-uniqueness, suggesting that a single technical production law does not exist. To organize the cointegrating models, we construct a fit-complexity envelope (E_{S1}) that identifies the optimal trade-off between statistical fit and parameter count:

$$E_{S1} = \{m \in \mathcal{A}_{S1}^F : \nexists m' \in \mathcal{A}_{S1}^F \text{ with } L(m') \leq L(m) \text{ and } K(m') \leq K(m)\}, \quad (17)$$

4.23 In this setup, $L(m) = -2 \log \hat{\ell}(m)$ is the model deviance (where lower values mean better fit) and $K(m)$ is the number of estimated parameters. A model lies on the envelope E_{S1} only if no other cointegrating model fits the data better while using fewer parameters.

4.24 We also define information-criterion neighborhoods ($\mathcal{F}_j^{(0.20)}$) to capture competitive specifications that sit slightly off the strict Pareto frontier:

$$\mathcal{F}_j^{(0.20)} = \{m \in \mathcal{A}_{S1}^F : IC_j(m) \leq Q_{0.20}(IC_j \mid \mathcal{A}_{S1}^F)\}, \quad (18)$$

4.25 Here, j indexes the criteria used to evaluate the models: the Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), Hannan-Quinn (HQ), and Bozdogan's Information Complexity Index (ICOMP) along with its robust variant (RICOMP). While AIC and BIC penalize the number of parameters, ICOMP and RICOMP penalize parameter interdependence and covariance complexity. This variety of criteria helps us see how different penalty structures select different long-run elasticity estimates across the specification space.

4.26 Each specification m yields a distinct estimate of the transformation elasticity $\hat{\theta}(m)$, which determines the slope of the capacity path. We construct the capacity utilization series for each model as:

$$\hat{\mu}_t(m) = y_t - [\hat{a}(m) + \hat{b}(m)t + \hat{\theta}(m)k_t]. \quad (19)$$

4.27 Comparing these utilization paths across the different model sets shows how modeling choices directly impact our capacity measurements. Selecting different cointegrating specifications shifts the estimated capacity ceiling, altering the calculated rate of capacity utilization.

4.28 Table 3 summarizes the structure of this single-equation specification search. We report the empirical results of this grid search in Subsection 4.5.

Table 3: Synthesis of Admissibility Specification Stress Tests

Stress-Test Component	Operational Layer	Screening/Organization Rule	Diagnostic Target
Cointegration admissibility	F/T-bounds sequence	$p_F(m) \leq \alpha, p_t(m) \leq \alpha$ (if defined) for $\alpha \in \{0.10, 0.05, 0.01\}$	Existence and significance of long-run level relationships
Specification invariance	Lag-order variation	Monitor parameter drift $\hat{\theta}(m)$ across combinatoric $p, q \in \{1, \dots, 5\}$	Stability and unique mapping of the long-run structural multiplier
Identification fragility	Deterministic cases & dummy structures	Track admissibility shifts across $c \in \{1, \dots, 5\}$ and $s \in \{s_0, \dots, s_3\}$	Sensitivity to deterministic components and structural trend assumptions
Information complexity & Pareto optimization	Fit-complexity envelope & IC neighborhoods	Pareto non-dominance in E_{S1} ; 20th percentile cutoff for $j \in \{\text{AIC, BIC, HQ, ICOMP, RCOMP}\}$	Organization of disciplined non-uniqueness under distinct information schedules
Capacity benchmark sensitivity	Utilization path fan $\hat{\mu}_t(m)$	Map structural parameter variance to latent residual series	Macroeconomic consequence of single-equation coefficient drift on normal profit estimation

Note: This synthesis matrix outlines the nested single-equation admissibility screens, information-theoretic optimization layers, and diagnostic targets within the stage $S1$ multi-model grid. Source: Author’s elaboration.

4.29 We extend this analysis to a joint multi-equation system in Stage $S2$. Transitioning to a system framework allows us to test whether bivariate cointegration holds when output and capital are both treated as endogenous, and whether establishing a stable long-run relationship requires including the rate of exploitation (e_t).

4.1.2 Stage 2 ($S2$): System Cointegration Test

4.30 Stage $S2$ tests whether the output–capital relation remains stable in a multi-equation system. If output and capital are cointegrated, a simple bivariate vector $X_t = (\ln Y_t, \ln K_t)'$ should exhibit a stable cointegrating relationship under Johansen’s rank tests. However, as we show in the results section, this bivariate system fails to cointegrate. We resolve this by introducing the logged rate of exploitation, $e_t = \ln(\pi_t/(1 - \pi_t))$, to form the trivariate vector $X_t = (\ln Y_t, \ln K_t, \ln e_t)'$. We

estimate this trivariate system using a standard Vector Error Correction Model (VECM):

$$\Delta X_t = \Pi X_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta X_{t-i} + \Phi D_t + \varepsilon_t, \quad (20)$$

4.31 In equation (20), Γ_i represents the short-run dynamics, D_t contains deterministic components and dummies, and Π is the long-run impact matrix. We factor this matrix as $\Pi = \alpha\beta'$, where β contains the long-run cointegrating vectors and α contains the adjustment speeds. A stable cointegrating relation exists only when the matrix has a reduced rank of $r = 1$, and the exploitation variable enters the cointegrating space β .

4.32 We test the stability of this system by estimating a grid of VECM specifications, varying the lag structures, trend configurations, and outlier dummies. For each target rank r , the tried specification space is:

$$\mathcal{G}_{S^2}^{(r)} = \{(p, C_{\text{det}}, s)\}, \quad (21)$$

4.33 Here, p represents the VAR lag orders (from 1 to 4), C_{det} indexes the four deterministic configurations, and s select the outlier dummies. In this system setup, C_0 assumes no constant or trend; C_1 restricts the constant to the cointegrating space; C_2 includes an unrestricted constant; and C_3 adds a linear trend restricted to the cointegrating space. These four cases are distinct from the single-equation PSS configurations. Permuting these choices yields 48 attempted models. We exclude the restricted constant models (C_1) because they systematically fail to converge, leaving 36 estimated models in our final grid.

4.2 Data, Measurement, and Stylized Facts

4.34 We use annual US corporate sector data from 1947 to 2011 ($T = 65$), sourced from the data companion in Shaikh (2016).³ Capacity utilization is the residual deviation of actual output from the cointegrated output–capital path. The level, timing, and persistence of our capacity path depend on these measurement choices. Table 4 summarizes the variables, their construction, and the measurement constraints that structure this replication.

4.35 To reconstruct the gross capital stock, we use the Generalized Perpetual Inventory Method (GPIM) following Shaikh (2016). This accounting framework calculates capital stock from nominal gross investment flows, annual depreciation rates, and an implicit deflator built from gross capital formation accounts. The recursion accumulates current-cost capital as:

$$K_t = IG_t + z_t^* \cdot K_{t-1}, \quad z_t^* \equiv (1 - z_t) \frac{p_t^K}{p_{t-1}^K}, \quad (22)$$

4.36 In this recursion, IG_t is nominal gross investment, z_t is the annual depreciation rate, and z_t^* refiles surviving capital to current replacement cost. We follow Shaikh’s three specific adjustments to construct this series: we replace the BEA 2011 infinite-lives assumption with BEA 1993 service

³Replication data and construction scripts are available at: <https://www.realecon.org/data>. While Shaikh specifies a common-deflator condition, he does not name the price index; we implement a GPIM-consistent price deflator to ensure reproducibility.

Table 4: Core replication variables: symbols, sources, and transformations

Symbol	Construction	Role	Measurement constraint
$GVAcorp_t$	NIPA Table 1.14 (L1) + imputed-interest adj. (Table 7.11)	Output flow	Classical surplus correction
$KGCcorp_t$	BEA Fixed Asset 6.1 (L9) + IRS SOI + GPIM recursion	Capital anchor	GPIM stock-flow reconstruction
p_t	pKN implicit price deflator, 2005=100	Common price basis	GPIM-consistent deflation
y_t	$\ln(VAcorp_t/p_t)$	Dependent variable	Inherits output and deflator choices
k_t	$\ln(KGCcorp_t/p_t)$	Regressor	Inherits GPIM depletion rates
D_{56}, D_{74}, D_{80}	Squared-residual diagnostics	Disturbance controls	Baseline outlier absorption

Note: Stage $S2$ introduces $\ln e_t$ (logged exploitation) to the trivariate state vector. Distributional variables enter only at the system-admissibility gate. Full series definitions, sources, and transformations are provided in Table 13 of Appendix B.

lives, anchor the initial values to 1925 benchmarks, and rescale the historical stock over 1925–1947 using the IRS corporate book-value index to account for scrapping during the Great Depression (Shaikh, 2016, Appendix 6.7). Our constructed series closely tracks the BEA official net stock with a 99.60% average ratio over 1947–2005, confirming that differences in levels reflect these methodological choices rather than measurement error. Appendix B.4 contains the full recursion inputs and depreciation rates.

4.37 We deflate both output and capital by the pKN price index (2005=100) to maintain stock-flow consistency. Standard BEA releases use quality-adjusted chain weights that alter the measured volume of investment across generations of capital, which breaks the physical accounting of capital accumulation. The pKN index avoids these quality-adjustment artifacts, ensuring that our output–capital relation reflects real productive capacity rather than statistical adjustments (Shaikh, 2016, Appendix 6.6). This approach satisfies the common-deflator condition and isolates real capacity dynamics. We detail this deflation rationale and compare it to quality-adjusted deflators in Appendix B.5.

4.38 We adjust corporate gross value added to correct for imputed financial intermediation services (FISIM) by subtracting net monetary interest paid and adding back imputed net interest adjustments from NIPA Table 7.11. We provide the algebraic derivation in Appendix B.2. We also include year dummies (D_{1956} , D_{1974} , D_{1980}) as shock absorbers to control for large historical breaks: the 1956 Eisenhower recession, the 1974 oil crisis, and the 1980 Volcker monetary contraction. These dummies prevent temporary shocks from distorting our long-run elasticity estimates. Since our sample size ($T = 65$) is relatively small, relying on asymptotic critical values could bias our cointegration tests. We address this limitation by computing finite-sample bounds calibrated to our exact sample length ($T = 65$) using stochastic simulations (Natsopoulos and Tzeremes, 2022).

4.39 These measurement choices are central to our empirical strategy. We use Shaikh’s GPIM-adjusted gross capital stock because our goal is to measure physical capacity in operation rather than the financial value of assets. While net capital stock measures are appropriate for calculating profit rates and asset valuation, gross capital stock closer matches the physical ability of capital to produce output.⁴ By using gross capital stock, we ensure that the estimated capacity benchmark is not distorted by accounting depreciation rules or market asset revaluations, which make net capital stock growth rates more volatile than physical capacity changes. The specific recursion parameters and service-lives adjustments used to construct this gross stock are detailed in Appendix B.4. Deflating both output and capital with the same price index further ensures that our estimates reflect real productive capacity rather than shifts in relative prices.

4.40 Between 1947 and 2011, post-war US capital accumulation systematically outpaced output and employment growth, driving a secular decline in the output-capital ratio that mirrors the broader OECD experience of Marx-biased technical change. Specifically, the real corporate gross capital stock grew at an average annual rate of 4.1%, while real output expanded at 3.0% and employment grew at only 1.3%. This structural decoupling is not unique to the United States; cross-country evidence confirms that labor productivity across advanced capitalist economies rises primarily through mechanization and capital-using, labor-saving technical change (Marquetti, 2003; Basu, 2010). This systematic depression of the output-capital ratio demonstrates that a fixed, technical relationship between capital and productive capacity is empirically invalid. Instead, the conversion of accumulated capital into active productive capacity is a historically conditioned process, fundamentally mediated by the choice of technique and distributive conflict.

4.41 We plot these post-war trends in Figures 2, 3, and 4. While output and capital grow together over the long run (Figure 2), their ratio is subject to long historical waves rather than remaining constant (Figure 3). As shown in Figure 4, output growth exhibits much higher volatility than capital accumulation. Economically, this lower volatility of capital stock growth is a standard macroeconomic stylized fact, but it is further pronounced here because we use gross capital stock (capital in operation) rather than net capital stock. Net capital stock growth rates are inherently more volatile because they incorporate short-run changes in financial book value, market revaluations, and accelerated accounting depreciation. In contrast, our GPIM-constructed gross capital stock remains insulated from these valuation shocks, reflecting the smoother physical survival of productive assets. Because capacity is a physical ceiling representing what capital can actually produce in use, gross capital stock is the theoretically correct anchor, whereas net capital stock measures the depreciated financial value of assets. Any estimated capacity utilization path must be consistent with these descriptive patterns; the parametric cointegration tests we run below must explain these raw historical movements rather than contradict them.

⁴This distinction is critical: financial valuation dominates profit-rate calculations, but capacity benchmarks require a measure of physical capital in use.

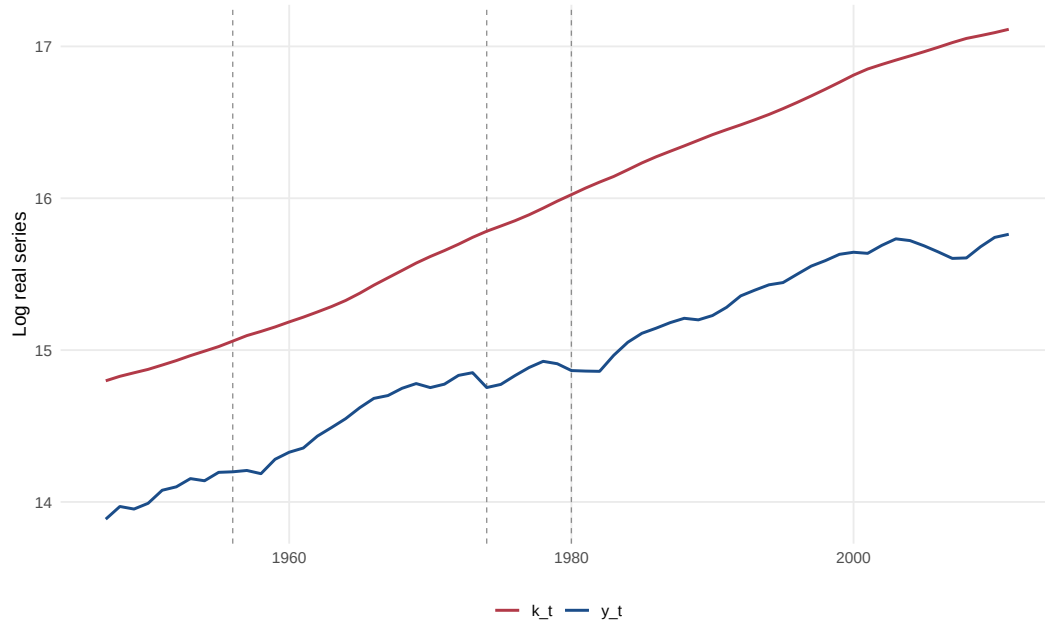


Figure 2: Log output (y_t) and log capital (k_t), 1947–2011. Vertical dashed lines mark 1956, 1974, and 1980.

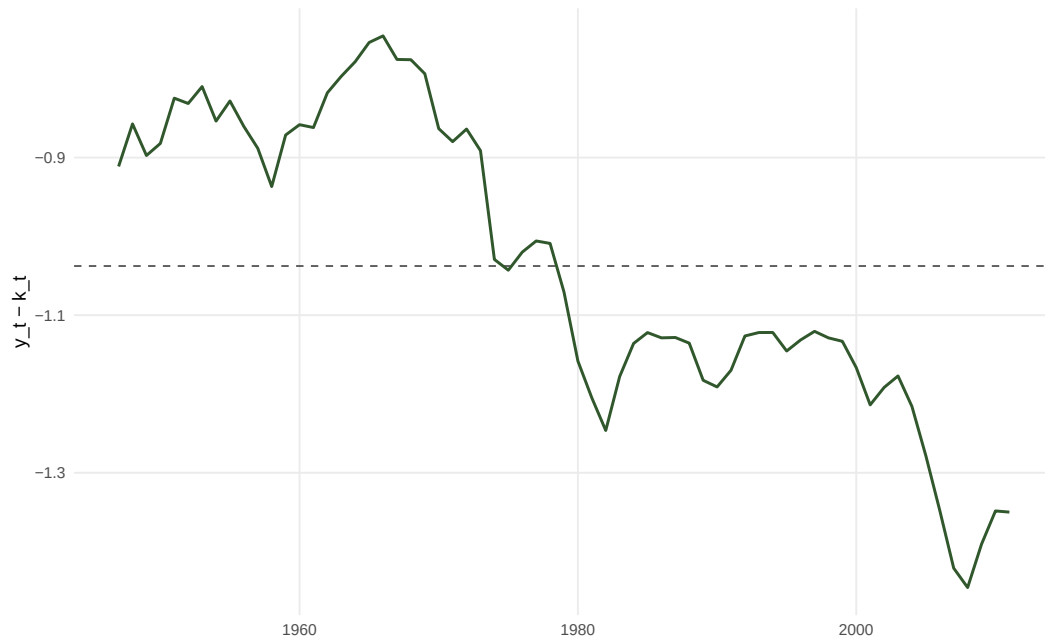


Figure 3: Output-capital ratio ($y_t - k_t$), 1947–2011. Horizontal line marks long-run mean.

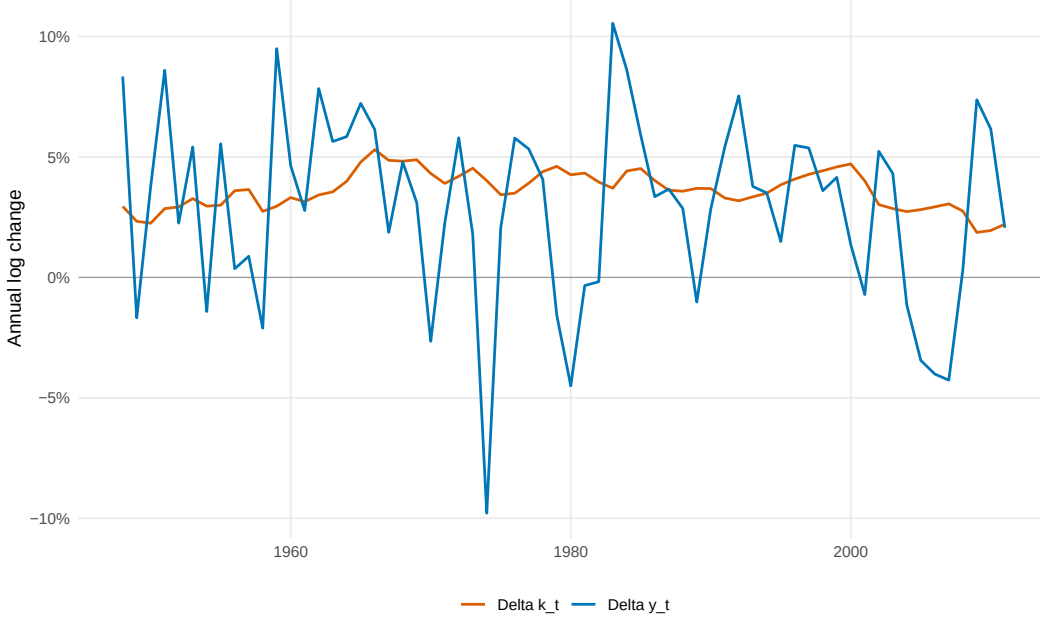


Figure 4: Annual growth rates of output (Δy_t), gross capital (Δk_t), and net capital (Δk_t^{net}), 1947–2011.

4.3 The Admissibility Strategy

4.42 Admissibility serves as a strict gatekeeping diagnostic to ensure that any capacity model represents a genuine long-run attractor rather than a spurious correlation. If output and capital fail to cointegrate, the residuals remain nonstationary, meaning the model merely fits noise and the derived capacity utilization series is a statistical artifact. We apply this filter sequentially: single-equation models in Stage S1 must pass the Wald F -bounds test and the t -bounds error-correction screen, while multi-equation VECMs in Stage S2 must satisfy cointegration rank, dynamic stability, and residual diagnostics. This sequential screening separates genuine structural relationships from researcher-driven specification artifacts.

4.43 Our first screen uses the joint F -bounds test to filter out specifications that fail to show a long-run relationship. A model is F -admissible if its Wald statistic rejects the null hypothesis of no cointegration at the 10% significance level. Formally, we define the F -admissible set $\mathcal{A}_{S1}^F$ as:

$$\mathcal{A}_{S1}^F = \{m \in \mathcal{G}_{S1} : p_F(m) \leq 0.10\}, \quad (23)$$

4.44 In this definition, $p_F(m)$ is the p -value of the Wald bounds test for model m , measuring the probability of the null hypothesis of no long-run relationship. If this probability is 10% or less, the specification enters our admissible pool. We discard any model that yields a Wald statistic below the lower critical bound or within the inconclusive interior band.

4.45 The secondary screen uses the t -bounds test to verify error-correction stability. Because the t -bounds statistic is only defined for deterministic Cases I, III, and V, we allow specifications using

Cases II and IV to bypass this test without penalty. The t -admissible set $\mathcal{T}_{S1}$ is:

$$\mathcal{T}_{S1} = \{m \in \mathcal{A}_{S1}^F : p_t(m) \leq 0.10 \text{ if defined}\}, \quad (24)$$

4.46 Here, $p_t(m)$ is the p -value for the one-sided t -test on the error-correction coefficient π_y . A significant negative coefficient confirms that output adjusts back toward the long-run capacity ceiling after a shock, rather than drifting away permanently. To identify specifications that pass both the Wald and error-correction tests at the stricter 5% level, we define the Strong Cointegration Space $\mathcal{C}_{S1}$ as:

$$\mathcal{C}_{S1} = \{m \in \mathcal{T}_{S1} : p_F(m) \leq 0.05 \text{ and } p_t(m) \leq 0.05\}, \quad (25)$$

4.47 The joint condition $p_F(m) \leq 0.05$ and $p_t(m) \leq 0.05$ requires both the level relationship and the speed-of-adjustment to be highly significant. To illustrate what is at stake in these screens, we run a "no-dummy" counterfactual specification where we omit the historical shock controls (D_{1956} , D_{1974} , D_{1980}). Omitting these dummies causes the residuals to remain nonstationary ($I(1)$) across all lag profiles, failing the cointegration bounds tests. This counterfactual proves that without controlling for major historical breaks, the output–capital relation appears spurious, demonstrating why these admissibility gates are necessary to filter out statistical artifacts.

4.48 In Stage S2, we evaluate the surviving models through a triple-gate system to ensure the estimates represent a stable multi-equation attractor rather than a spurious correlation. First, the Cointegration Rank Gate requires the Johansen Trace test to confirm a single cointegrating vector ($r = 1$) at the 5% level, showing that the variables share a common long-run path. Second, the Dynamic Stability Gate requires all eigenvalues of the VECM companion matrix to lie strictly inside the unit circle (excluding the unit roots of the common trends), filtering out explosive models where temporary shocks cause output to diverge infinitely. Third, the Residual Diagnostic Gate requires the model residuals to pass tests for serial correlation and heteroscedasticity at the 5% level, ensuring our statistical inference is unbiased. In the trivariate family, the rank condition $r = 1$ checks whether output and capital cointegrate once we include income distribution. We rank the surviving VECMs using their log-likelihood values to identify the best-fitting systems.

4.3.1 Cross-Stage Synthesis

4.4 Stage 0 (S0): Single-Equation Reconstruction

4.49 Stage S0 confirms that Shaikh’s baseline capacity path is numerically reproducible, but this recovery hinges on undocumented data choices rather than a unique structural law. The divergence between the published multiplier of 0.66 and our reconstructed estimate of 0.72 arises from undocumented choices in data construction and lag selection. Reconstructing this benchmark required tracing exact price deflators and data vintages through archive spreadsheets, demonstrating that forensic replication is an active econometric reconstruction rather than a passive mirror. Consequently, the closest reconstruction yields an elasticity of $\hat{\theta} = 0.72$, which barely clears the 10

4.50 This divergence between the published 0.66 and our reconstructed 0.72 is our first major finding. This gap is not a calculation error; it arises from undocumented choices in the original data construction. Reconstructing his benchmark required tracing the exact price deflators and data

vintages through archive spreadsheets, where alternative deflator selections shift the coefficient. This shows that forensic replication is an active econometric reconstruction rather than a passive mirror. The closest reconstruction yields an elasticity of $\hat{\theta} = 0.72$ from an ARDL(2,4) specification, which matches the model selected by the Bayesian Information Criterion (BIC). However, this ARDL(2,4) model yields an F -bounds statistic of 3.349, which barely clears the 10% significance threshold ($p = 0.099$). To address this cointegration fragility, we select the AIC-chosen ARDL(4,3) specification as our focal model. This model yields a more stable estimate of $\hat{\theta} = 0.75$ and cointegrates at the 5% level ($F = 5.250$, $p = 0.023$). We report the parameters for both models in Table 5.

4.51 We report the bounds test statistics and finite-sample critical values for both specifications in Table 5. By using stochastically simulated finite-sample distributions rather than asymptotic tables, we ensure that our cointegration thresholds are calibrated to our sample size. In the faithful ARDL(2,4) reconstruction, the F -bounds statistic of 3.349 barely clears the 10% critical value threshold of 3.333 ($p = 0.099$). In contrast, our focal ARDL(4,3) model yields a highly significant F -bounds statistic of 5.250 ($p = 0.023$) and a t -bounds statistic of -3.125 ($p = 0.015$), confirming that the relationship cointegrates robustly.

Table 5: Stage S0 ARDL Bounds Evidence at $\alpha = 0.10$ (90% Confidence Intervals) with Small-Sample Critical Values

Model	Test	PSS Case	Statistic	I(0) Bound	I(1) Bound	Exact-Sample p	Verdict
ARDL(2,4)	F -bounds	Case 1	3.349	2.466	3.333	0.099	Reject H_0
ARDL(2,4)	t -bounds	Case 1	-2.507	-1.602	-2.259	0.062	Reject H_0
ARDL(4,3)	F -bounds	Case 1	5.250	2.486	3.338	0.023	Reject H_0
ARDL(4,3)	t -bounds	Case 1	-3.125	-1.621	-2.270	0.015	Reject H_0

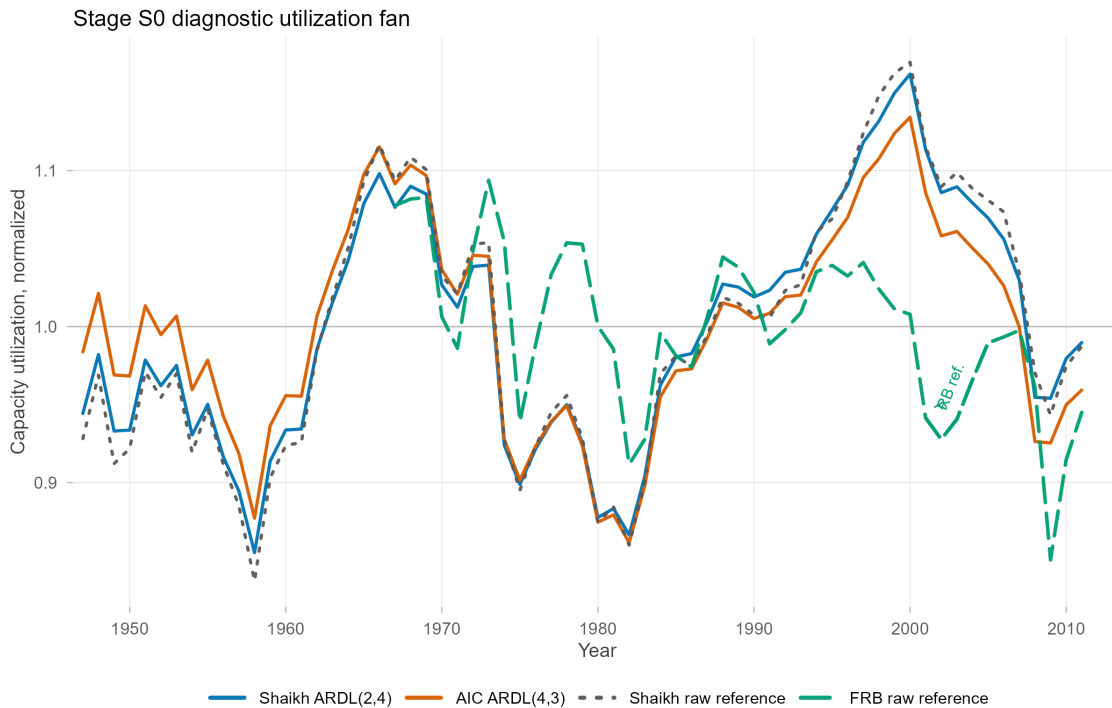
Note: The null hypothesis assumes no long-run level relation. The F -bounds statistic supplies the primary admissibility diagnostic. The t -bounds statistic is reported as a stricter secondary diagnostic where the deterministic structure permits. Case 1 denotes the no-constant/no-trend PSS deterministic configuration. Rejection is assessed at $\alpha = 0.10$. Source: Author's elaboration.

4.52 The error-correction estimates for the focal ARDL(4,3) model show a slow adjustment process back to the capacity path. The speed of adjustment coefficient $\hat{\pi}_y$ is -0.127 (SE 0.038, $t = -3.34$, $p < 0.01$), meaning the economy corrects only 12.7% of its deviation from the capacity ceiling each year. The negative coefficients on our historical dummies (D_{1956} , D_{1974} , D_{1980}) are all statistically significant, capturing permanent step-shifts in capital productivity. These shifts correspond to major institutional disruptions: the 1956 Eisenhower defense cuts, the 1974 OPEC oil crisis, and the 1980 Volcker monetary contraction.

4.53 A small change in the estimated transformation elasticity θ from 0.72 to 0.75 alters the constructed capacity utilization series by up to 5 percentage points. This sensitivity occurs because the elasticity acts as the exponent of the capital stock in our capacity function, so minor changes in the parameter shift the entire capacity ceiling. This amplification shows why point-estimate replication is insufficient: small differences in specification translate into large differences in historical interpretation. We illustrate this sensitivity in Figure 5, which plots the capacity utilization paths generated by these two models, alongside Shaikh's published series and the Federal Reserve Board (FRB) manufacturing index. The gap between the reconstructed ARDL(2,4) path ($\hat{\theta} = 0.72$) and

the ARDL(4,3) focal path ($\hat{\theta} = 0.75$) highlights how minor differences in lag structures shift the utilization series, showing that single-equation capacity paths are fragile benchmarks for policy analysis.

Figure 5: S0 Diagnostic Utilization Fan



Note: The fan maps the variance of constructed utilization paths given competing ARDL specifications. Source: Author's elaboration.

4.54 Stage S0 confirms that we can numerically reproduce a capacity path similar to Shaikh's. However, recovering a single point estimate does not guarantee that the parameter is unique or stable across other specifications. To test this parameter's sensitivity, we transition to Stage S1 and estimate a broader grid of single-equation models.

4.5 Stage 1 (S1): Admissible-Specification Stress Test Results

4.55 Opening the specification space to a 500-model ARDL grid shatters the illusion of a unique, fixed technical multiplier. Instead of converging on a single structural law, the admissible estimates scatter across a wide parameter space: only 102 of the 500 specifications survive the 10

4.56 To illustrate these results, we compare two focal models that represent the opposite edges of our specification space. The first is the model selected by the standard AIC, BIC, HQ, and ICOMP criteria: an ARDL(3,3) specification under PSS Case I (no constant, no trend) with the 1974 dummy, yielding a high elasticity estimate of $\hat{\theta} = 0.92$ (SE 0.04). The second is the model selected by the robust RICOMP criterion: a parsimonious ARDL(1,2) specification under PSS Case II (restricted constant) with no dummies, yielding a much lower elasticity estimate of $\hat{\theta} = 0.65$ (SE 0.07). We report the parameters and diagnostics for these two models in Table 6.

Table 6: Stage S1 Focal Single-Equation Estimations

Parameter	AIC/BIC/HQ/ICOMP Focal: ARDL(3,3)			RICOMP Focal: ARDL(1,2)		
	Estimate	SE	<i>t</i> -stat	Estimate	SE	<i>t</i> -stat
$\hat{\theta}$ (elasticity)	0.920	0.040	23.00***	0.651	0.070	9.29***
a (intercept)	— (Case 1)			0.584	0.402	1.45
D_{1974}	-0.038	0.017	-2.24**	— (s_0)		
F -bounds statistic	8.120 (Case 1) [Reject ***]			3.269 (Case 2) [Reject **]		
t -bounds statistic	-4.520 (Case 1) [Reject ***]			-3.125 (Case 2) [Reject **]		
$1 - \sum \alpha_i$ (denominator)	0.240			0.091		
Breusch-Godfrey LM (p -value)	0.48			0.39		
CUSUM/CUSUMSQ	Stable			Stable		

Note: Small-sample critical values are simulated for $T = 65$. Significance: *** $p < 0.01$, ** $p < 0.05$. Source: Author's elaboration.

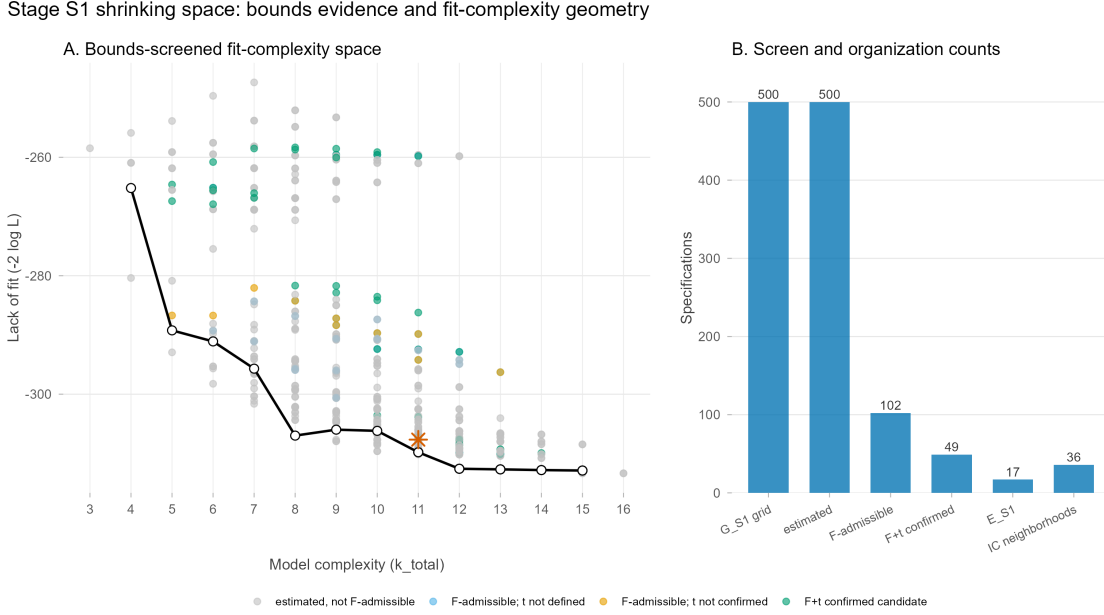
Table 7: S1 Shrinking Specification Space Counts

Layer	Count
Full grid	500
Estimated successfully	500
F-admissible at 10%	102
F+t confirmed at 10%	49
F-admissible at 5%	62
F+t confirmed at 5%	28
F-admissible at 1%	13
F+t confirmed at 1%	3
E_{S1}	17
IC-neighborhood union	36

Note: This table tracks the survival rate of the 500 estimated single-equation ARDL models through the nested cointegration admissibility gates. Source: Author's elaboration.

4.57 We visualize this screening process in Figure 6. Panel A maps all 500 specifications across the fit-complexity space, color-coded by their cointegration status. Panel B tracks the number of surviving specifications at each diagnostic layer. This sequence reinforces the methodological rule that we must screen models for cointegration before comparing their fit or complexity.

Figure 6: S1 Shrinking Specification Space



Source: Author's elaboration.

4.58 Within the cointegrating set, we construct a fit-complexity envelope (E_{S1}) containing 17 optimal specifications. This envelope represents the Pareto frontier where we cannot improve a model's fit without adding parameters, and we cannot simplify it without worsening its fit. The envelope serves as a geometric summary of the best-performing single-equation alternatives rather than a single preferred model.

4.59 The heavy reliance on dummy variables matches the historical shocks of the post-war US economy. The shock years mark major disruptions: the 1956 Eisenhower recession (following defense cuts after the Korean War), the 1974 OPEC oil embargo (inducing stagflation), and the 1980 Volcker interest rate shock (triggering a credit crunch). Bivariate models that omit these dummy variables yield nonstationary residuals because these crises act as permanent shifts in the output–capital relation. Restoring a stationary error term requires these step-shift dummies to control for the structural drops in capital productivity. Specifically, the negative dummy coefficients capture the permanent dampening of capital productivity (Y/K) caused by these crises: the post-Korean War manufacturing slowdown in 1956, the OPEC price-profit squeeze forcing capacity to idle in 1974, and the Volcker shock pushing interest rates to 20% and permanently lowering capacity utilization in 1980.

4.60 We define information-criterion neighborhoods ($\mathcal{F}_j^{(0.20)}$) to organize the cointegrating specifications. For each criterion, the neighborhood collects the top 20% of cointegrating models, resulting in a union of 36 unique specifications. Table 8 reports the parameter ranges across these neighborhoods, illustrating how different model-selection criteria organize the specification space.

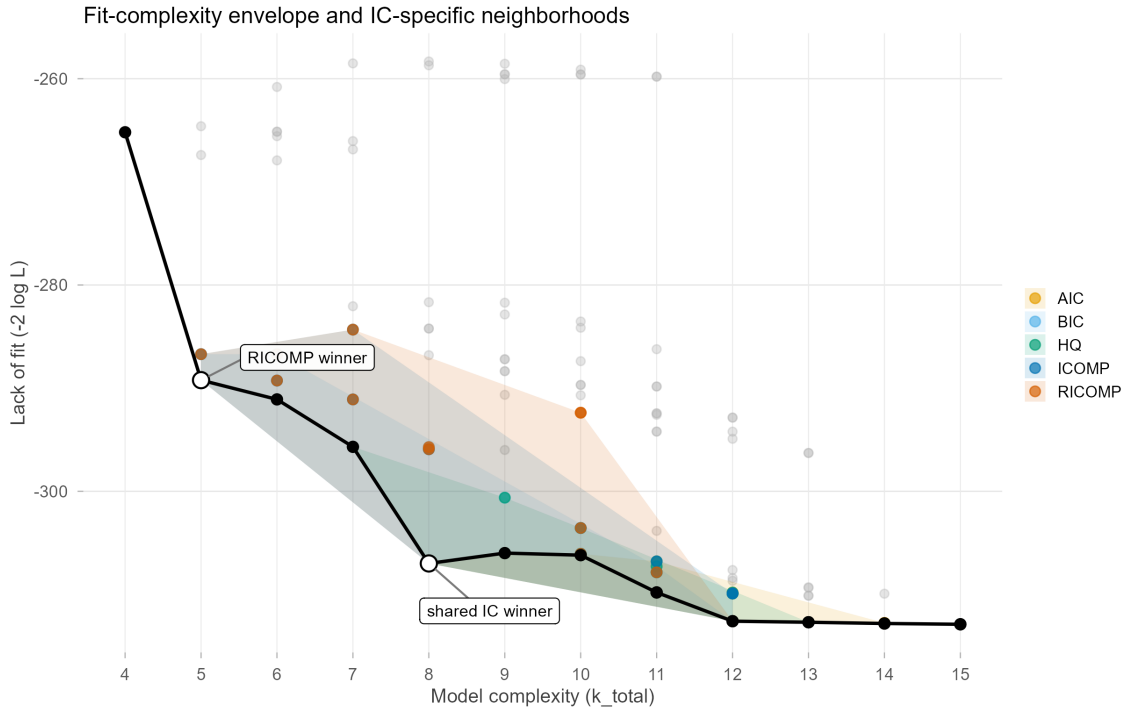
Table 8: S1 IC-Specific Neighborhood Summaries

Criterion	Specs	Min $\hat{\theta}$	Mean $\hat{\theta}$	Median $\hat{\theta}$	Max $\hat{\theta}$
AIC	22	0.84	0.88	0.86	0.93
BIC	21	0.65	0.91	0.86	2.20
HQ	21	0.75	0.94	0.86	2.20
ICOMP	22	0.65	0.84	0.86	0.95
RICOMP	22	0.65	0.83	0.85	0.95

Note: The table reports the summary statistics for the long-run parameter $\hat{\theta}$ within the bottom 20th percentile of each information criterion. Estimates are rounded to two decimal places for consistency. Source: Author's elaboration.

4.61 Figure 7 maps these criterion neighborhoods against the fit-complexity envelope. The solid line represents the Pareto frontier (E_{S1}), while the shaded areas show the neighborhoods for each criterion. This map highlights the spatial divergence between the parsimonious models selected by RICOMP and the more complex models selected by AIC and BIC.

Figure 7: S1 Fit-Complexity Envelope and IC-Specific Neighborhoods



Source: Author's elaboration.

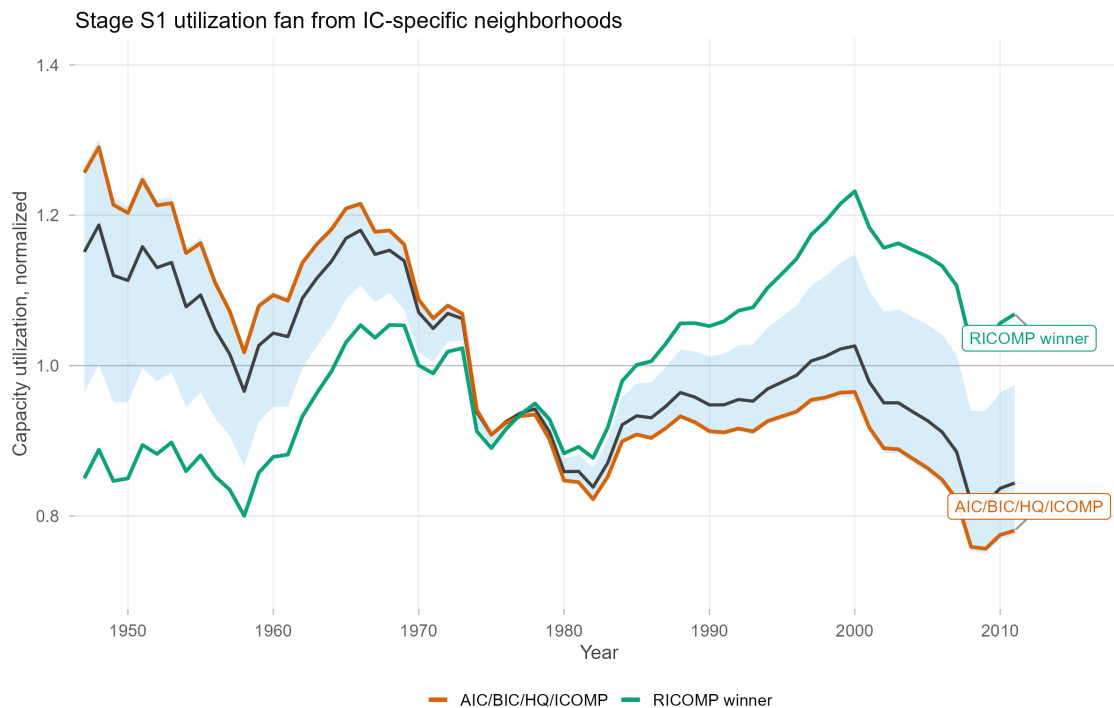
4.62 This criteria divergence reveals that different information-theoretic penalties locate very different regions of the specification space. AIC and BIC penalize the number of parameters, selecting the complex ARDL(3,3) model with $\hat{\theta} = 0.92$. In contrast, ICOMP and RICOMP penalize parameter interdependence and covariance complexity, selecting the parsimonious ARDL(1,2) model with $\hat{\theta} = 0.65$. This parameter variation has significant theoretical stakes. While no-trend configurations

(PSS Cases I, II, and III) yield stable elasticities below unity ($\hat{\theta} \in [0.65, 0.95]$) that support the overaccumulation hypothesis, trend-containing configurations (Cases IV and V) yield elasticities averaging 1.36 and reaching up to 2.20. An elasticity $\theta > 1$ implies that productive capacity expands faster than capital accumulation in the long run, violating basic physical constraints. We discard these trend-containing models not because they fail statistical bounds tests, but because they imply implausible long-run capacity dynamics.

4.63 Figure 8 plots the utilization paths generated by these different cointegrating models. The shaded fan shows the variance of constructed capacity utilization, while the divergent paths of the AIC and RICOMP winners highlight how model selection choices mechanically alter our historical interpretation of capacity realization.

4.64 To illustrate the physical stakes of these estimates, we run a counterfactual model imposing the balanced-growth constraint ($\theta = 1.0$), forcing capacity to grow in lockstep with capital. Under this constraint, the constructed capacity utilization series drifts to economically absurd levels, falling continuously to under 35% for decades and exceeding 105% during cyclical peaks. A corporate sector cannot operate at 35% capacity for thirty years without widespread bankruptcy, nor can it operate above 100% capacity indefinitely. This physical absurdity shows that the balanced-growth assumption is incompatible with the post-war US historical experience, confirming that we must estimate the elasticity θ freely to obtain a viable capacity utilization series.

Figure 8: S1 Diagnostic Utilization Fan from IC-Specific Neighborhoods



Source: Author's elaboration.

4.65 Stage S1 shows that while many single-equation specifications pass cointegration tests, they yield very different estimates of the transformation elasticity ($\hat{\theta}$). Because these models do not converge to a single parameter, any capacity path derived from a single-equation model remains fragile. In Subsection 4.6, we address this limitation by testing whether the output–capital relation survives in a joint multi-equation VECM system, and whether stability requires us to include the rate of exploitation.

4.6 Stage 2 (S_2): System-Level VECM Replication

4.66 Stripping the output-capital relation down to a bivariate VECM exposes its fundamental econometric fragility. Across every lag profile and deterministic configuration, the bivariate system fails to cointegrate, leaving residuals trapped in a nonstationary ($I(1)$) drift. This breakdown demonstrates that capital and output cannot sustain a stable long-run trajectory in isolation; the bivariate relation is simply spurious. Establishing a genuine long-run capacity benchmark therefore requires abandoning the insulated technical assumption and explicitly conditioning the system on the macroeconomic distribution of income.

4.67 To resolve this system-level instability, we expand the state vector to a trivariate system $X_t = (\ln Y_t, \ln K_t, \ln e_t)'$, where $e_t = \pi_t/(1 - \pi_t)$ is the rate of exploitation, measured as the corporate profit-share-to-wage-share ratio. The rate of exploitation proxies the balance of class power that dictates shop-floor discipline, work intensity, and the physical conversion of capital. When labor is strong, firms may deliberately idle capacity to protect profit margins; conversely, weakened labor unions and "just-in-time" speedups allow firms to extract more output from the same physical stock. Thus, the survival of the output-capital relation only when income distribution enters the cointegrating space indicates that capacity utilization is not a neutral technical law, but a contingent outcome of decentralized accumulation and the balance of class power.

4.68 We summarize these system-level outcomes in Table 9. All 36 bivariate specifications fail to cointegrate. In contrast, the trivariate family yields six admissible models when we include the rate of exploitation and our historical dummy variables. These results prove that the output–capital relation achieves system stability only when we account for income distribution.

Table 9: S2 System Admissibility Outcomes

System	Rank	Estimated	Admissible	Interpretation
Bivariate ($\ln Y, \ln K$)	$r = 1$	36	0	Output–capital relation is not self-sufficient
Trivariate ($\ln Y, \ln K, \ln e$)	$r = 1$	36	6	Restricted survival with logged exploitation
Trivariate ($\ln Y, \ln K, \ln e$)	$r = 2$	36	0	No admissible two-vector system

Source: Author’s elaboration.

4.69 To ground the VECM results, we select the VAR(2) specification under Johansen Case 2 (C_2 : restricted constant), h_2 dummies, and rank $r = 1$ as our focal model. The model passes all three admissibility gates. We verify cointegration rank using the Johansen Trace test: we reject

the null hypothesis of no cointegration ($r = 0$) at the 5% level (Trace statistic $32.28 > 31.52$), and we fail to reject the null of a single cointegrating vector ($r \leq 1$) (Trace statistic $10.68 < 17.95$). We confirm dynamic stability through the VECM companion matrix eigenvalues: the moduli are 1.0, 1.0, 0.82, 0.52, 0.52, 0.49, all strictly bounded by the unit circle (except the two unit roots). Residual diagnostics verify that the system is free from residual pathology: the Portmanteau(12) statistic is 100.70 ($p = 0.275$), the Jarque-Bera statistic is 5.72 ($p = 0.456$), and the ARCH-LM(4) statistic is 150.03 ($p = 0.348$).⁵ We report the estimated cointegrating vector (β) and loading vector (α) in the following paragraphs.

4.70 Normalizing the cointegrating vector on output yields:

$$\ln Y_t - 0.727 \ln K_t + 20.022 \ln e_t - c_0 \sim I(0) \quad (26)$$

The estimated capital elasticity $\hat{\theta} = 0.727$ matches our single-equation results. The large standard error (4.852) reflects the multicollinearity between the capital stock and the rate of exploitation over the post-war period. The coefficient on the rate of exploitation is highly significant ($\hat{\beta}_e = 20.022$, SE 2.536, $t = 7.90$), showing a strong level relationship between productive capacity and income distribution.

4.71 The adjustment speeds (α) show a structural asymmetry in the system. Capital accumulation is weakly exogenous ($\alpha_k = 0.000$, $t = 0.92$), meaning that investment decisions do not respond to long-run deviations from the output–capital relation; instead, investment drifts independently of the long-run equilibrium. The output equation adjusts slowly but significantly ($\alpha_y = -0.005$, $t = -2.00$). The primary channel of error correction is the rate of exploitation ($\alpha_e = -0.019$, $t = -4.25$). Economically, this means that when output exceeds the capacity ceiling (relative to capital and exploitation), the rate of exploitation falls to restore system stability. This finding of weak exogeneity in capital accumulation ($\alpha_k = 0.000$) provides empirical support for the Sraffian position in the classical-heterodox debate. By showing that investment does not adjust to error-correction signals of capacity utilization disequilibrium, the data reject the Neo-Kaleckian view of investment adapting endogenously to demand-induced utilization gaps. However, because our long-run coefficient represents a non-unitary transformation elasticity ($\hat{\theta} = 0.727 < 1.0$), this Sraffian exogeneity is accompanied by structural overaccumulation. Under this interpretation, the secular instability of unbalanced growth and the exogeneity of capital accumulation act as counteracting forces that may cancel each other out in the long run. This potential canceling obscures the resolution of the debate between Sraffian and Neo-Kaleckian models, a result that is invisible when working under a balanced-growth baseline with a fixed output–capital ratio. We report the short-run dynamics (Γ_1) and deterministic dummy coefficients (Φ) in Table 10.

⁵The Breusch-Godfrey LM(4) statistic of 60.81 ($p = 0.006$) indicates residual serial correlation, a common small-sample VECM limitation that we document for transparency.

Table 10: Focal VECM Short-Run Dynamics and Dummies

Regressor / Dummy	$\Delta \ln Y_t$ Equation	$\Delta \ln K_t$ Equation	$\Delta \ln e_t$ Equation
$\Delta \ln Y_{t-1}$	0.199 (0.150)	0.047** (0.023)	-0.798** (0.314)
$\Delta \ln K_{t-1}$	-1.015 (0.564)	0.812*** (0.087)	-4.023*** (1.183)
$\Delta \ln e_{t-1}$	-0.023 (0.067)	0.015 (0.010)	-0.069 (0.140)
D_{1956} (dummy)	-0.005 (0.015)	0.002 (0.002)	-0.062** (0.029)
D_{1974} (dummy)	-0.038** (0.017)	0.001 (0.002)	-0.078** (0.032)
D_{1980} (dummy)	0.001 (0.016)	-0.003 (0.002)	-0.014 (0.031)

Note: Standard errors are in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$. Source: Author's elaboration.

4.72 The historical dummy variables are essential for establishing system-level cointegration. Cointegration holds only when we include the step-shift dummies to control for structural breaks, preventing permanent shifts in the mean of the residuals from appearing as unit roots. Under the trend-containing Johansen Case 3 (C_3) configurations, the models yield invalid parameter estimates, including negative ($\hat{\theta} = -0.81$) or explosive ($\hat{\theta} = 11.31$) elasticities. In these configurations, the linear trend absorbs the long-run growth of output, forcing the capital coefficient to fit short-run cyclical noise. We therefore reject these trend-containing models. The economically viable models are restricted to no-trend Cases C_0 and C_2 , which yield stable, positive elasticity estimates ranging from 0.73 to 0.97 (Table 11).

Table 11: S2 Retained Trivariate Specifications

Lag Order (p)	Deterministic Case (C)	Historical Dummy	Rank (r)	Cointegrating Coefficient ($\hat{\theta}$)
1	C_0	h_2	1	0.91
1	C_2	h_2	1	0.97
1	C_3	h_2	1	-0.81
2	C_0	h_2	1	1.19
2	C_2	h_2	1	0.73
2	C_3	h_2	1	11.31

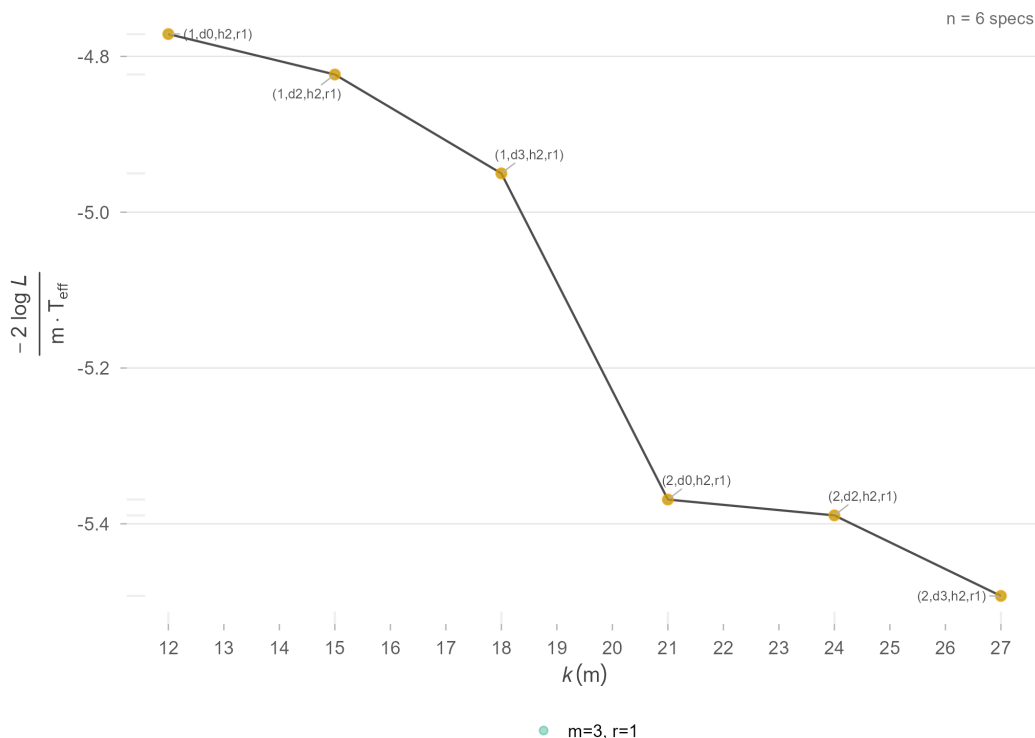
Source: Author's elaboration.

4.73 To illustrate the role of historical controls, we run a "no-dummy" counterfactual where we omit the dummy variables. When we leave out these shock controls, the VECM fails to reject the null hypothesis of no cointegration under both Johansen trace and maximum eigenvalue tests. The residuals remain nonstationary ($I(1)$), proving that the estimated relationship is spurious. Output, capital, and exploitation drift apart permanently without a restoring force. This counterfactual demonstrates that the output-capital relation is not an institution-free physical law; its stability depends on controlling for major post-war shocks like the 1974 oil crisis and the 1980 Volcker credit contraction.

4.74 We plot the system-level fit-complexity frontier in Figure 9. Bivariate systems fail to cointegrate, and adding variables beyond the rate of exploitation does not improve system-level

identification. This confirms that the trivariate system is the minimum specification necessary for econometric admissibility.

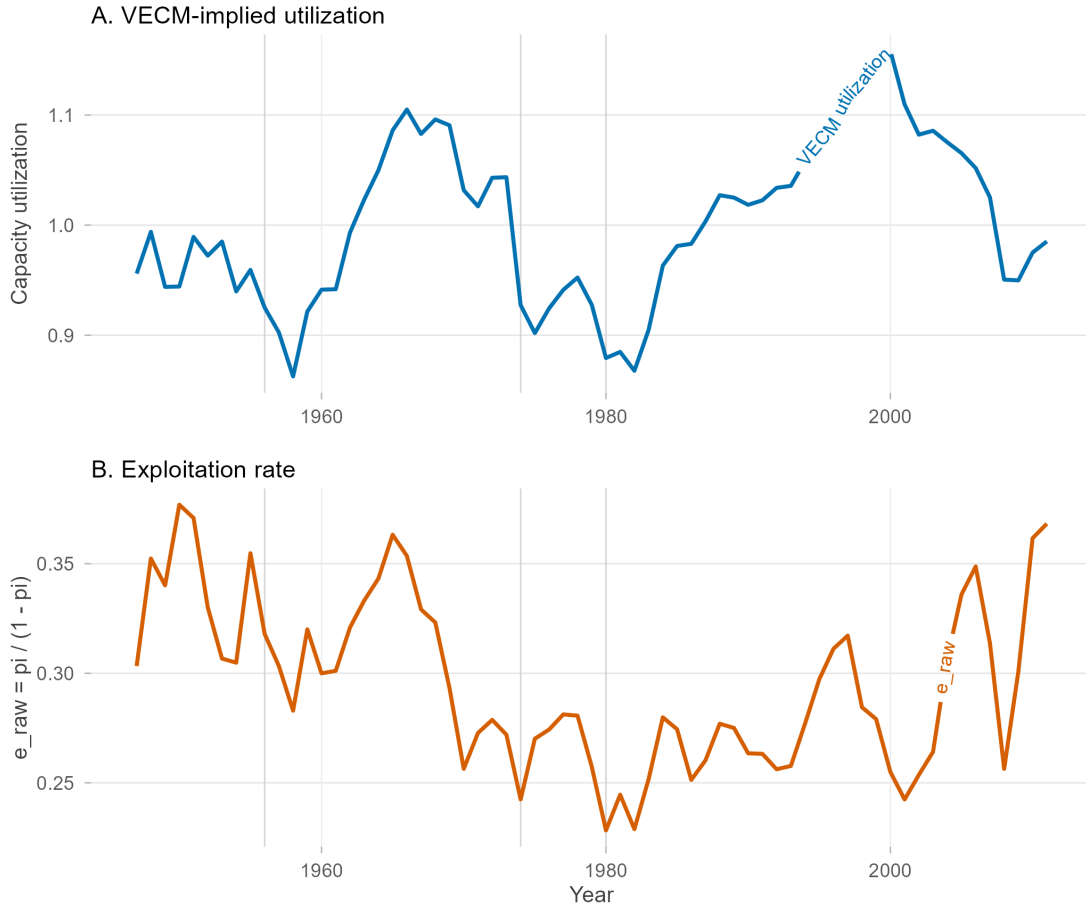
Figure 9: S2 Pooled System Frontier



Source: Author's elaboration.

4.75 Figure 10 plots the constructed rate of capacity utilization alongside the rate of exploitation. In the United States, the rate of exploitation peaked in the mid-1960s, declined during the stagflation of the 1970s, and rose steadily from 1983 to 2011 as wage growth lagged behind productivity. This trajectory reveals a structural link between capacity utilization and the functional distribution of income, showing that the long-run realization of capacity is conditioned by class struggle.

Figure 10: S2 Focal Utilization and Exploitation Diptych



Source: Author's elaboration.

4.76 Kurz's (1986) classical choice-of-technique argument provides the theoretical mechanism for this econometric requirement of including a distributive conflict variable into the VECM. Changes in income distribution alter the relative cost-effectiveness of production methods, forcing firms to switch techniques and adjust capacity utilization to protect profit margins. Because firms respond to these distributional shifts by altering their choice of technique, the output-capital ratio (Y/K) undergoes structural breaks that a purely bivariate model cannot absorb. Omitting the rate of exploitation therefore forces the error term to capture these unmodeled distributional shifts, generating the spurious drift observed in Stage S2. Ultimately, the survival of the output-capital relation only when income distribution enters the cointegrating space indicates that capacity utilization is not a neutral technical law, but a historically contingent outcome of decentralized accumulation and distributive conflict.

4.7 Cross-Stage Econometric Synthesis

4.77 The three-stage replication defines the strict historical and institutional boundary conditions of Shaikh's capacity measurement strategy. While the baseline single-equation coefficient is numerically

reproducible (Stage S0), it remains structurally non-unique and highly sensitive to information-criterion choices (Stage S1). Most critically, the standalone output-capital relation fails system-level cointegration, surviving exclusively within a trivariate VECM conditioned on income distribution and historical shock controls (Stage S2). The long-run conversion of capital to capacity is therefore not a fixed engineering multiplier. It is an institutional reality, fundamentally shaped by systemic crises and the ongoing balance of power between wages and profits.

Table 12: Cross-Stage Synthesis of the Cointegration Architecture

Stage	Outcome	Interpretive Bound
S0	Approximate single-equation reproducibility	Shaikh’s method is reproducible, but not exactly
S1	Non-empty admissible ARDL grid; non-unique retained benchmark	Model selection is driven by information-criterion choice
S2 Bivariate	36 estimated, 0 admissible	Output–capital system is not self-sufficient
S2 Trivariate Rank One	36 estimated, 6 admissible	Restricted survival with logged exploitation and h_2 controls
S2 Trivariate Rank Two	36 estimated, 0 admissible	No retained dual-vector system

Source: Author’s elaboration.

4.78 The value of the estimated transformation elasticity ($\hat{\theta}$) dictates both the level of our capacity utilization path and its long-run trend. Since utilization is the residual difference between actual output and fitted capacity, the elasticity determines how much of the long-run capital trend we subtract from output. A lower elasticity estimate (such as RICOMP’s 0.65) forces the residual to absorb structural drift, generating a utilization path characterized by long historical waves and slow error correction. A higher elasticity estimate (such as AIC’s 0.92) strips the capital trend from the residual, resulting in a highly stationary utilization series that mirrors short-run business cycles. This variation highlights how technical specification choices shape our historical interpretation of capacity realization.

4.79 System-level VECM estimations resolve the single-equation ambiguity by exposing the historical and institutional conditions needed for stability. All six surviving trivariate models require the historical dummy vector (D_{1956} , D_{1974} , D_{1980}). This pattern shows that the outlier controls in Shaikh’s baseline are not arbitrary adjustments; they are necessary to capture a stable cointegrating space. Cointegration holds only when we control for these major historical crises, which stabilize the long-run relationship between capital accumulation and output.

4.80 The collapse of the standalone output–capital relation stems from the dual role of capital accumulation. Investment demand expands current output in the short run, while the physical capital stock expands productive capacity in the long run. The parameter θ is forced to mediate these conflicting forces. In a single-equation framework, this entanglement creates parameter instability. The system-level VECM results prove that capacity utilization is not a simple engineering multiplier.

In this replication, a stable capacity path depends on both historical shock controls and income distribution. Productive capacity must be measured as a historically conditioned political-economy object rather than a neutral technical ceiling.

5 Conclusion

5.1 The standalone output-capital relation cannot sustain a stable capacity benchmark; cointegration emerges only when the rate of exploitation enters the system. This critical replication of Shaikh (2016) accounting-based methodology demonstrates that the bivariate VECM fails across all lag profiles and deterministic configurations, without any evidence of cointegration between output and capital stocks. Instead, cointegration holds only in a trivariate VECM conditioned on income distribution and historical shock controls. Once trend-containing specifications are excluded as economically inadmissible, the retained models yield a transformation elasticity below unity ($\hat{\theta} < 1.0$), placing the post-war US corporate sector in a structural overaccumulation regime where capital accumulation outpaces capacity formation in the long-run.

5.2 The non-unitary transformation elasticity imposes a secular downward drag on the rate of profit that capitalists can only counteract by intensifying exploitation or idling capacity. Within the classical-Marxian decomposition $r = (\Pi/Y) \cdot \mu \cdot (Y^p/K)$, a transformation elasticity below unity forces the capital-capacity ratio (Y^p/K) into a persistent decline as capital accumulates. The speed of adjustment of the system suggests that capital accumulation is weakly exogenous ($\alpha_k \approx 0$), while the rate of exploitation serves as the primary error-correction channel restoring long-run reproduction. This weak exogeneity of investment supports the Sraffian position over the Neo-Kaleckian demand-endogenous view. Yet the structural overaccumulation ($\hat{\theta} < 1.0$) introduces a counteracting dynamic: the secular instability of unbalanced growth and the weak exogeneity of investment may cancel each other out, obscuring the resolution of this debate. This theoretical impasse reflects the shared balanced-growth assumption underlying both frameworks, which forces a fixed output-capital ratio onto a historically shifting accumulation process. Ultimately, the capacity ceiling is not a neutral engineering constraint but a contingent outcome of decentralized accumulation, shaped by the historical balance of class power and the distributive conflicts of the capitalist labor process.

5.3 The single-sector bounding and the time-invariant parameterization of θ represent the two primary analytical limitations of this chapter. By abstracting from inter-sectoral disproportionalities, capacity transfers, and uneven development across departments of production, the single-sector framework isolates the aggregate capital-to-capacity relation at the cost of tracing how sectoral imbalances spill over to affect the general rate of profit. Similarly, treating θ as a constant over the 1947–2011 period forces the model to average across historically distinct choice-of-technique regimes and social structures of accumulation. While external demand components—military spending, state deficits, and consumer debt expansion—may stabilize the overaccumulation tendency, these transmission mechanisms remain hypotheses rather than findings directly estimated here.

5.4 The empirical failure of the standalone output-capital relation establishes a clear theoretical imperative: capacity measurement must integrate workplace-level conflict *ex ante* rather than appending it *ex post*. The immediate next step is to endogenize the transformation elasticity θ

as a time-varying parameter driven by class struggle and the choice of technique, tracing how the capital-capacity conversion shifts across historical US accumulation regimes. Building on this closed-economy foundation for the center, the dissertation must subsequently confront the structural realities of the periphery. In peripheral economies, the productive ceiling is not merely bounded by domestic distributive conflict, but is structurally truncated by the Balance of Payments (BoP) constraint. For these economies, potential capacity is subordinated to the external viability of the accumulation process; the imperative to generate sufficient foreign exchange to finance essential imports and service external debt acts as a hard, macroeconomic ceiling on domestic expansion. The next analytical layer will therefore relax the single-sector bounding to incorporate open-economy dynamics, estimating a capacity utilization rate for peripheral countries that explicitly conditions the output-capital relation on these external constraints. This upgrade will reveal whether chronic idle capacity in the periphery is driven by domestic overaccumulation or by the structural necessity of maintaining external equilibrium, ultimately demonstrating that the capacity ceiling in the Global South is a geopolitical and distributive construct rather than a neutral technical limit.

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A Derivation of Accumulation Dynamics under Unbalanced Growth

This appendix provides the derivation of the dynamical equations summarized in §3.3 and §3.4. All results follow from three components: the accounting identity $\hat{k} + \delta = \phi^p B$, the unbalanced growth closure $g_{Y^p} = \theta g_K$, and the investment-savings closure $\phi^p = s$.

A.1 The Unbalanced Growth Closure: Derivation of the Bernoulli ODE

The accounting identity under the investment-savings closure is:

$$\hat{k} + \delta = s B. \quad (27)$$

Differentiating with respect to time yields:

$$\frac{d\hat{k}}{dt} = s \dot{B}. \quad (28)$$

Expressing $\dot{B}$ in terms of the growth rate of capital productivity ($\hat{b}$):

$$\dot{B} = B \hat{b}. \quad (29)$$

Substituting this into equation (28) and utilizing the accounting identity $sB = \hat{k} + \delta$ results in the acceleration identity:

$$\frac{d\hat{k}}{dt} = (\hat{k} + \delta) \hat{b}. \quad (30)$$

Recall that capital productivity is $B \equiv Y^p/K$, which in growth rates implies $\hat{b} = \hat{y}^p - \hat{k}$. Under the unbalanced growth closure $g_{Y^p} = \theta g_K$ (or $\hat{y}^p = \theta \hat{k}$ in growth rate notation), this gives the growth rate of capital productivity as $\hat{b} = (\theta - 1) \hat{k}$. Substituting this into equation (30) provides the acceleration dynamic:

$$\boxed{\frac{d\hat{k}}{dt} = (\theta - 1) \hat{k} (\hat{k} + \delta)}, \quad \text{where } \hat{k} \text{ and } \delta \text{ are expressed in consistent temporal and physical units.} \quad (31)$$

This is a Bernoulli equation of order 2 in $\hat{k}$, where the nonlinearity arises from the product $\hat{k}(\hat{k} + \delta)$.

A.2 Analytical Solution

Expanding equation (31) gives:

$$\frac{d\hat{k}}{dt} = (\theta - 1) \hat{k}^2 + (\theta - 1) \delta \hat{k}. \quad (32)$$

Applying the Bernoulli substitution $v \equiv \hat{k}^{-1}$ with $\dot{v} = -\hat{k}^{-2} \dot{\hat{k}}$ transforms the equation into a first-order linear ODE:

$$\dot{v} + (\theta - 1) \delta v = -(\theta - 1). \quad (33)$$

The solution for $v(t)$, given initial conditions $v_0 = \hat{k}_0^{-1}$, is:

$$v(t) = \left(v_0 + \frac{1}{\delta} \right) e^{-(\theta-1)\delta t} - \frac{1}{\delta}. \quad (34)$$

Inverting $v(t)$ yields the explicit path for the net accumulation rate:

$$\hat{k}(t) = \frac{1}{\left(\frac{1}{\hat{k}_0} + \frac{1}{\delta} \right) e^{-(\theta-1)\delta t} - \frac{1}{\delta}}. \quad (35)$$

A.3 Equilibria and Stability Analysis

The equilibria $\hat{k}^*$ are defined by $(\theta - 1)\hat{k}(\hat{k} + \delta) = 0$. For $\theta \neq 1$, the roots are $\hat{k}_1^* = 0$ (stationary net capital) and $\hat{k}_2^* = -\delta$ (zero gross investment). Linearizing around $\hat{k}^* = 0$ via the Jacobian $f'(\hat{k}) = (\theta - 1)(2\hat{k} + \delta)$ yields:

- Over-accumulation ($\theta < 1$): $f'(0) < 0$ — the stagnation equilibrium is locally stable.
- Under-accumulation ($\theta > 1$): $f'(0) > 0$ — the stagnation equilibrium is unstable.

A.4 The Trend-Stabilized Closure: Derivation of the Quadratic ODE

Under the trend-stabilized closure $\hat{y}^p = \theta\hat{k} + b$, the growth rate of capital productivity is:

$$\hat{b} = (\theta - 1)\hat{k} + b. \quad (36)$$

Substituting this into the acceleration identity (equation 30) generates a quadratic ODE:

$$\frac{d\hat{k}}{dt} = (\hat{k} + \delta) [(\theta - 1)\hat{k} + b]. \quad (37)$$

The equilibria are the roots of $(\hat{k} + \delta)[(\theta - 1)\hat{k} + b] = 0$:

$$\hat{k}_1^* = -\delta, \quad \hat{k}_2^* = \frac{b}{1 - \theta}. \quad (38)$$

The Jacobian $g'(\hat{k}) = 2(\theta - 1)\hat{k} + (\theta - 1)\delta + b$ evaluated at $\hat{k}_2^*$ confirms local stability for the empirically relevant case ($\theta < 1$ and $b > 0$): the interior root $b/(1 - \theta)$ acts as a strictly positive attractor, preventing stagnation. As $b \rightarrow 0$, the trend-stabilized closure collapses to the unbalanced growth attractor at $\hat{k}^* = 0$.

B Data Construction and Diagnostic Support

B.1 Series Glossary

Table 13 lists the core replication variables, their sources, and transformation rules. All series derive from Shaikh’s (2016) data companions via Real Economic Analysis.⁶

Table 13: Replication series: symbols, sources, and transformations

Symbol / Code	Variable	Source (table, line)	Transformation
$GVAcorp_t$	Corporate gross value added (corrected)	NIPA 1.14 (L1) + imputed-interest adj. (Table 7.11)	nominal
$KGCcorp_t$	Corporate gross capital stock (GPIM-adjusted)	Fixed Asset 6.1 (L9); IRS Statistics of Income; GPIM recursion	nominal
$KNCcorp_t$	Corporate net capital stock (GPIM-adjusted)	same as above	nominal
$KNCcorpbea_t$	Corporate net capital stock (BEA 2011 official)	Fixed Asset 6.1 (L9)	nominal
p_t	Common deflator	pKN index, 2005=100	$(KNCcorp/KNRcorp)*100$
y_t	Real log output	constructed	$\ln(VAcorp_t/p_t)$
k_t	Real log capital	constructed	$\ln(KGCcorp_t/p_t)$
$Profshcorp_t$	Corporate profit share (corrected)	NIPA 1.14 (L8, L1–L2) + imputed-interest adj.	ratio

Note: All BEA and NIPA data downloaded in the replication vintage. Shaikh’s original series used data downloaded in 2011 (Shaikh, 2016, Appendix 6.7, fn. 1); discrepancies attributable to subsequent BEA revisions are possible and noted where material.

Source: [Real Economic Analysis Data Tables](#)

B.2 Imputed-Interest Correction

The NIPA corporate value added figure includes imputed financial intermediation services as a cost to non-financial corporations. The corporate imputed-interest adjustment removes this component:

$$CorpImpIntAdj_t \equiv -BankMonIntPaid_t - CorpNFNetImpIntPaid_t, \quad (39)$$

where $BankMonIntPaid_t$ is NIPA Table 7.11 (lines 4+44+73 minus lines 28+52+91) and $CorpNFNetImpIntPaid_t$ is NIPA Table 7.11 (line 74 minus line 53). In 2009 the adjustment raises corporate net operating surplus by approximately 10 %. Figure 11 displays the corrected versus uncorrected series and their ratio.

B.3 Corrected Corporate Output: $GVAcorp_t$

$$GVAcorp_t \equiv GVAcorpnipa_t + CorpImpIntAdj_t, \quad (40)$$

where $GVAcorpnipa_t$ is NIPA Table 1.14 (line 1) and $CorpImpIntAdj_t$ is defined in equation (39).

⁶Replication data and construction scripts are available at Real Economic Analysis: <https://www.realecon.org/data>.

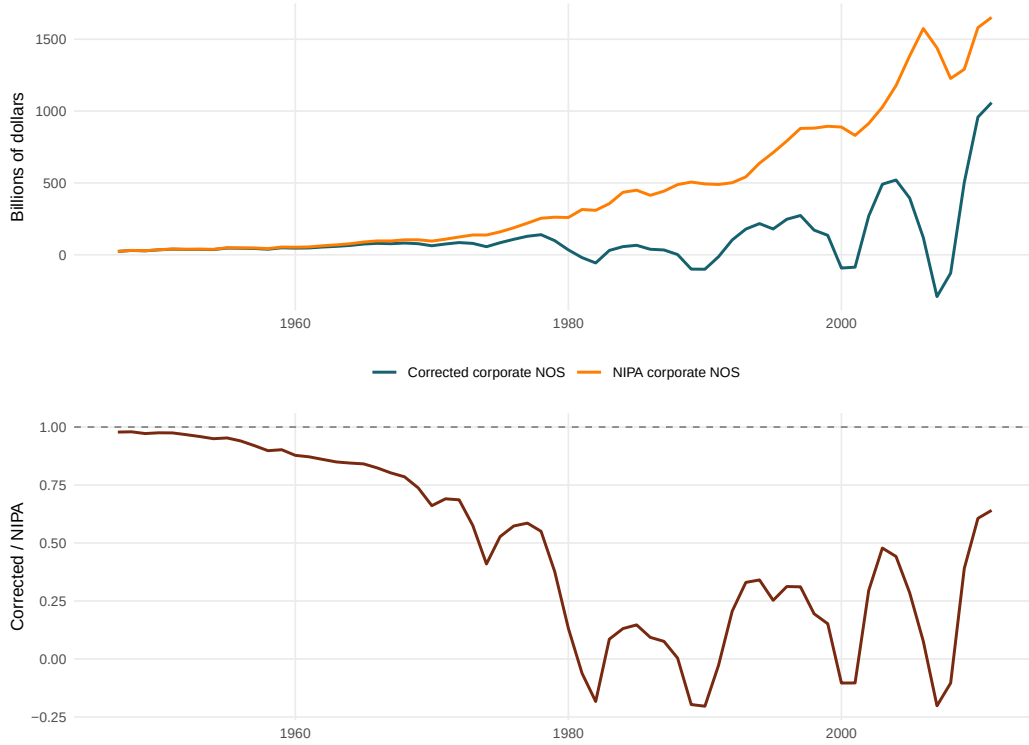


Figure 11: Imputed interest adjustment impact: corrected corporate NOS versus NIPA corporate NOS (1947–2011). Ratio shows magnitude of adjustment.

B.4 Reconstructed Corporate Capital Stock: $KGC_{corp,t}$

GPIM accumulation rule. The current-cost capital stock is accumulated as:

$$K_t = IG_t + z_t^* \cdot K_{t-1}, \quad z_t^* \equiv (1 - z_t) \frac{p_t^K}{p_{t-1}^K}, \quad (41)$$

where IG_t is nominal gross investment, z_t is the period depreciation rate, and z_t^* is the survival-and-revaluation factor that reflates the surviving stock to current replacement cost.

Three adjustments (applied before the recursion).

1. **Depletion rates.** BEA 1993 finite service lives replace the BEA 2011 infinite-lives assumption, implying higher depreciation and lower retirement rates consistent with physical exhaustion.
2. **Initial values.** BEA 1993 initial values anchored to 1925 replace the BEA 2011 starting values. The 1993 values are lower, producing faster accumulation growth and convergence to the official path by approximately 1977.
3. **Great Depression correction.** The IRS corporate book value index ([of the Census, 1975](#), Series V 115, pp. 924–926) is applied over 1925–1947, rescaling the BEA historical stock by the IRS/BEA ratio each year. The GPIM recursion resumes from 1947. Net effect: $KNC_{corp,t}$ is approximately 28 % lower in 1947 than the BEA 2011 official measure.

Inventories. IRS corporate balance sheet inventory ratios are extrapolated over 1947–1989 via a linear fit to the 1990–2011 IRS/BEA ratio, rescaled to the adjusted GPIM stocks, and added to $KGCcorp_t$. Source: IRS Statistics of Income 1926–2011; FRB Flow of Funds FL105015205.A for non-financial corporate inventories in current cost.

Accuracy. The average ratio of the GPIM approximation to the BEA current-cost stock over 1947–2005 is 99.60 %.

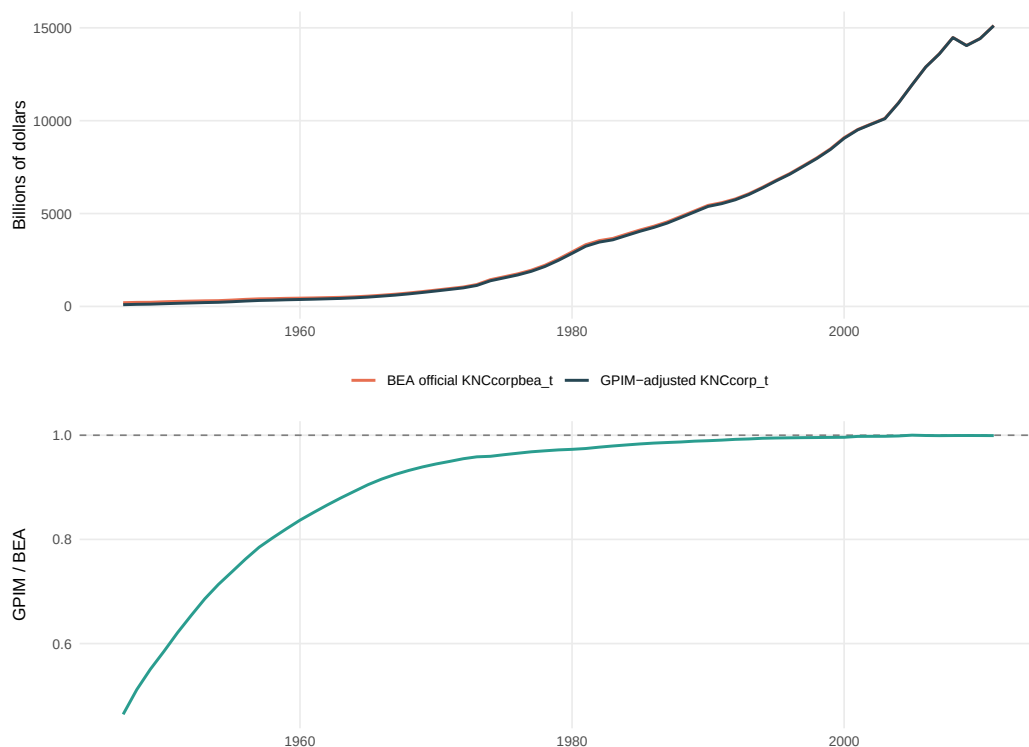


Figure 12: GPIM-adjusted net capital stock versus BEA 2011 official series (1947–2011). Ratio plot shows convergence by 1977.

B.5 Common Deflator: p_t

The replication uses the pKN implicit price deflator based at $100 = 2005$, estimated under GPIM rules. Current BEA releases employ quality-adjusted chain weights that adjust for quality vintage, breaking stock-flow consistency for accumulation accounting. The pKN index preserves stock-flow consistency, ensuring the output–capital relation reflects real productive dynamics rather than quality-adjustment artifacts (Shaikh, 2016, Appendix 6.6, Sec. II).

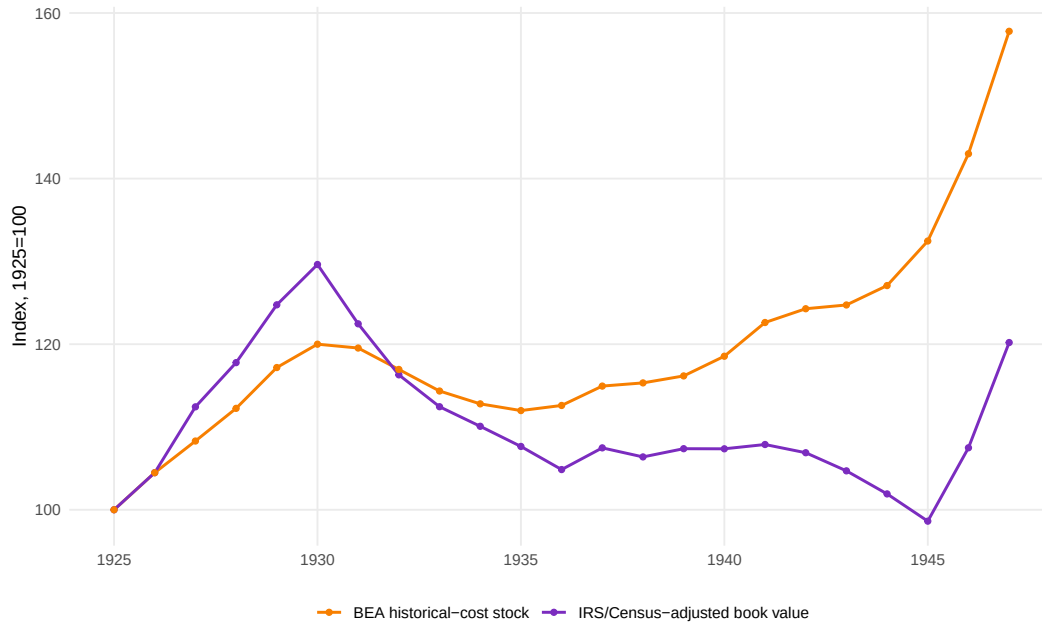


Figure 13: IRS corporate book value index versus BEA historical stock (1925–1947). Both indexed to 100 in 1925.

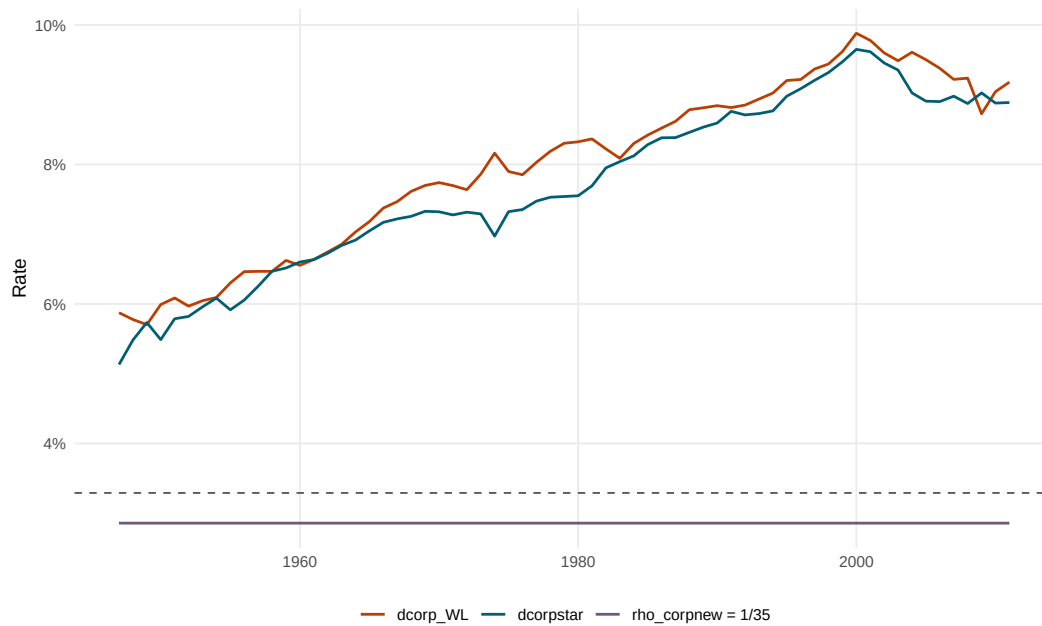


Figure 14: Depreciation and retirement rates under BEA 1993 finite service lives. Horizontal line marks critical value (3.29%).

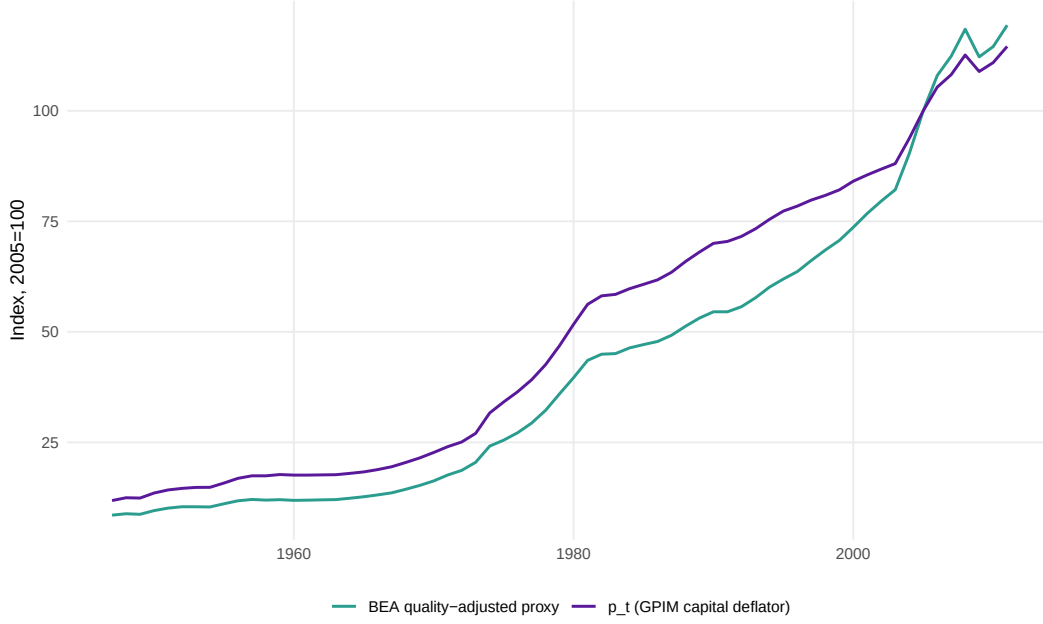


Figure 15: Common deflator index: GPIM-consistent pKN versus BEA quality-adjusted proxy (2005=100).

B.6 Corrected Corporate Profit Share: Profshcorp_t

The same imputed-interest adjustment applied to GVAcorp_t is applied to both the numerator and denominator of the profit share:

$$\text{NOScorp}_t \equiv \text{NOScorpnipa}_t + \text{CorpImpIntAdj}_t, \quad (42)$$

$$\text{VAcorp}_t \equiv \text{VAcorpnipa}_t + \text{CorpImpIntAdj}_t, \quad (43)$$

$$\text{Profshcorp}_t \equiv \frac{\text{NOScorp}_t}{\text{VAcorp}_t}, \quad (44)$$

where NOScorpnipa_t is NIPA Table 1.14 (line 8) and VAcorpnipa_t is NIPA Table 1.14 (line 1 minus line 2).

B.7 Descriptive Statistics and Unit Root Tests

Table 14 reports summary statistics for the core series. Table 15 presents unit root test results (ADF, Phillips-Perron) for y_t and k_t in levels and first differences.

Table 14: Core series summary statistics (1947–2011)

Variable	N	Mean	Median	Std Dev	Min	Max	Skewness	Kurtosis
y_t	65	14.9281	14.8653	0.5674	13.8866	15.7628	−0.1438	1.8398
k_t	65	15.9658	15.9810	0.7313	14.7979	17.1127	−0.0206	1.6872
Δy_t	64	0.0293	0.0355	0.0391	−0.0979	0.1055	−0.6331	3.4760
Δk_t	64	0.0362	0.0361	0.0078	0.0186	0.0530	−0.1127	2.4022

p_t	65	50.3486	46.8900	32.8159	11.8262	114.5539	0.3955	1.8063
D_{56}	65	0.0154	0.0000	0.1240	0.0000	1.0000	7.8750	63.0156
D_{74}	65	0.0154	0.0000	0.1240	0.0000	1.0000	7.8750	63.0156
D_{80}	65	0.0154	0.0000	0.1240	0.0000	1.0000	7.8750	63.0156

Note: Source: Real Economic Analysis (<https://www.realecon.org/data>), Shaikh (2016) data companions. Variables constructed per Table 13 of Appendix B. Dummy variables are impulse indicators with mean $1/65 \approx 0.0154$.

Table 15: Unit root tests for core series (1947–2011)

Variable	Form	Test	Specification	Statistic	Lag/BW	Critical values			Reject at 5%
						1%	5%	10%	
y_t	level	ADF	none	3.0027	1	-2.60	-1.95	-1.61	no
y_t	level	PP	intercept	-1.3280	3	-3.53	-2.91	-2.59	no
y_t	level	PP	trend	-2.1341	3	-4.11	-3.48	-3.17	no
y_t	Δ	ADF	intercept	-5.0966	1	-3.51	-2.89	-2.58	yes
y_t	Δ	PP	intercept	-5.9637	3	-3.54	-2.91	-2.59	yes
y_t	Δ	PP	trend	-5.9460	3	-4.11	-3.48	-3.17	yes
k_t	level	ADF	none	1.1552	4	-2.60	-1.95	-1.61	no
k_t	level	PP	intercept	-0.3863	3	-3.53	-2.91	-2.59	no
k_t	level	PP	trend	-1.2114	3	-4.11	-3.48	-3.17	no
k_t	Δ	ADF	intercept	-2.0670	3	-3.51	-2.89	-2.58	no
k_t	Δ	PP	intercept	-1.9027	3	-3.54	-2.91	-2.59	no
k_t	Δ	PP	trend	-1.8431	3	-4.11	-3.48	-3.17	no

Note: ADF: Augmented Dickey–Fuller test with max lag 6 and AIC selection. PP: Phillips–Perron test with Newey–West bandwidth. "none" = no deterministic terms; "intercept" = constant only; "trend" = constant + linear trend. PP "none" specification not computed. Rejection at 5% indicates stationarity under that specification.

Figures showing the log levels of output and capital, the output–capital ratio, and their annual growth rates are presented in the main text (see Section 4.2, Figures 2, 3, and 4).

B.8 ARDL Specification Search and Bounds Testing

Table 16 displays the full specification grid ($p, q = 1-6$; Cases I–V) with information criteria, diagnostic tests, and recovered long-run coefficients. Asterisks mark case-specific AIC/BIC minima.

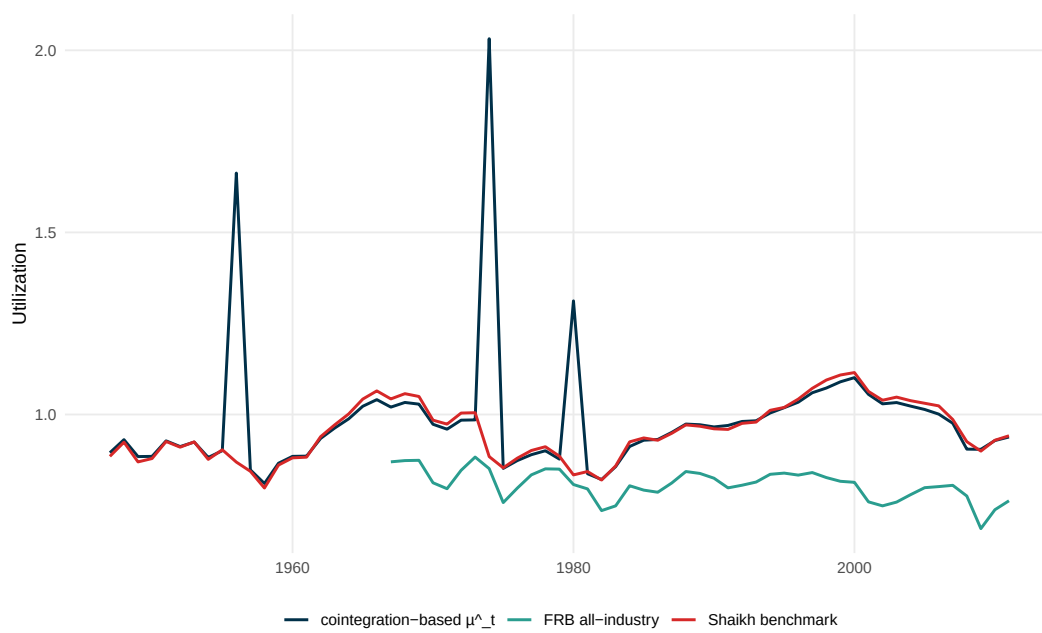


Figure 16: Capacity utilization comparison: Shaikh (cointegration-based) versus FRB survey measure, 1947–2011. Descriptive comparison only.

Table 16: ARDL specification search: information criteria and diagnostics (1947–2011)

p	q	Case	AIC	SBC	T_{eff}	$\hat{\theta}$	LM(1)	p -val	LM(4)	p -val	Best
1	1	I	−237.63	−222.52	64	0.8529	3.2749	0.070	5.0893	0.278	
1	1	II	−243.09	−225.82	64	0.6989	3.7150	0.054	6.9003	0.141	
1	1	III	−243.09	−225.82	64	0.6989	3.7150	0.054	6.9003	0.141	
1	1	IV	−243.09	−223.66	64	−0.5048	3.4486	0.063	6.5385	0.162	
1	1	V	−243.09	−223.66	64	−0.5048	3.4486	0.063	6.5385	0.162	
1	2	I	−239.33	−222.19	63	0.7273	7.0198	0.008	11.1845	0.025	
1	2	II	−242.69	−223.41	63	0.7110	7.1117	0.008	11.8692	0.018	
1	2	III	−242.69	−223.41	63	0.7110	7.1117	0.008	11.8692	0.018	
1	2	IV	−241.90	−220.47	63	−0.4718	7.0457	0.008	11.2449	0.024	
1	2	V	−241.90	−220.47	63	−0.4718	7.0457	0.008	11.2449	0.024	
1	3	I	−244.02	−224.88	62	0.8967	11.3499	0.001	14.1599	0.007	
1	3	II	−249.33	−228.05	62	0.6695	10.3605	0.001	12.7048	0.013	
1	3	III	−249.33	−228.05	62	0.6695	10.3605	0.001	12.7048	0.013	
1	3	IV	−249.62	−226.22	62	−0.8749	10.3560	0.001	11.9612	0.018	*
1	3	V	−249.62	−226.22	62	−0.8749	10.3560	0.001	11.9612	0.018	*
2	3	I	−250.91	−229.63	62	0.9301	5.4621	0.019	9.3503	0.053	
2	3	II	−262.79	−239.40	62	0.7023	0.5641	0.453	3.5633	0.468	*
2	3	III	−262.79	−239.40	62	0.7023	0.5641	0.453	3.5633	0.468	*
2	3	IV	−264.33	−238.81	62	−0.4836	0.2849	0.594	2.6102	0.625	*
2	3	V	−264.33	−238.81	62	−0.4836	0.2849	0.594	2.6102	0.625	*
4	5	I	−253.33	−224.01	60	0.7899	0.0461	0.830	7.6617	0.105	*
4	5	II	−257.75	−226.33	60	0.7102	0.2628	0.608	6.6363	0.156	
4	5	III	−257.75	−226.33	60	0.7102	0.2628	0.608	6.6363	0.156	
4	5	IV	−257.11	−223.60	60	−0.4117	0.2120	0.645	6.5141	0.164	
4	5	V	−257.11	−223.60	60	−0.4117	0.2120	0.645	6.5141	0.164	

Note: ARDL(p, q) specifications estimated with impulse dummies D_{56}, D_{74}, D_{80} . AIC/SBC: Akaike/Schwarz information criteria (lower = better). T_{eff} : effective sample size after lag construction. $\hat{\theta}$: long-run output–capital coefficient recovered from ECM parameters. LM(1)/LM(4): Breusch–Godfrey serial correlation tests (null = no autocorrelation). Asterisk (*) marks best specification by AIC/SBC within each deterministic case. Full grid ($p, q = 1$ –6; Cases I–V) available in replication code.

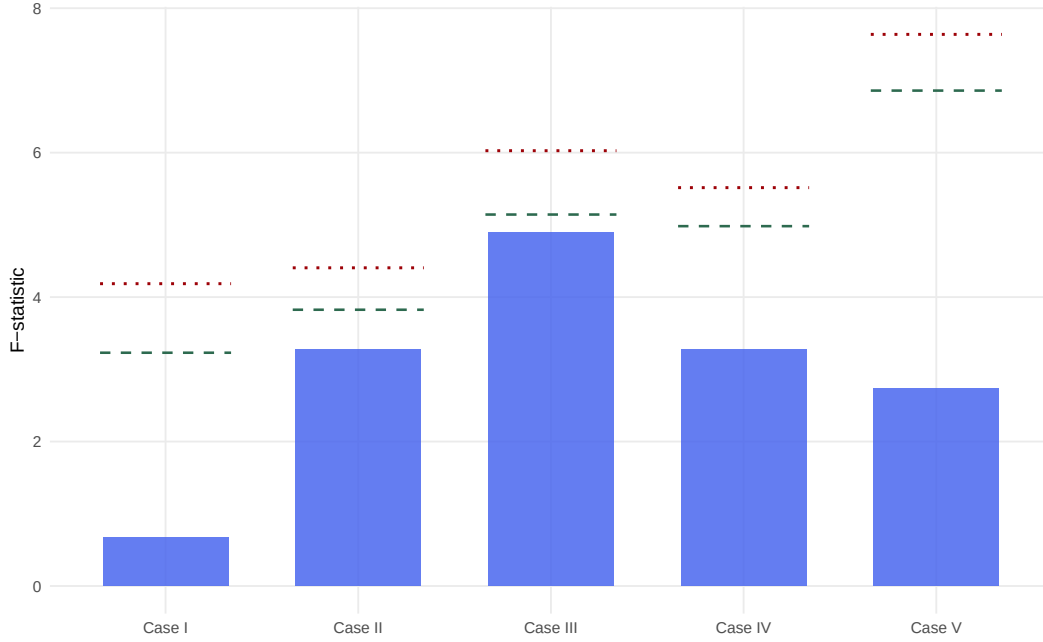


Figure 17: F -statistic across deterministic Cases I–V. Horizontal lines mark lower ($I(0)$) and upper ($I(1)$) critical bounds at 5% significance.

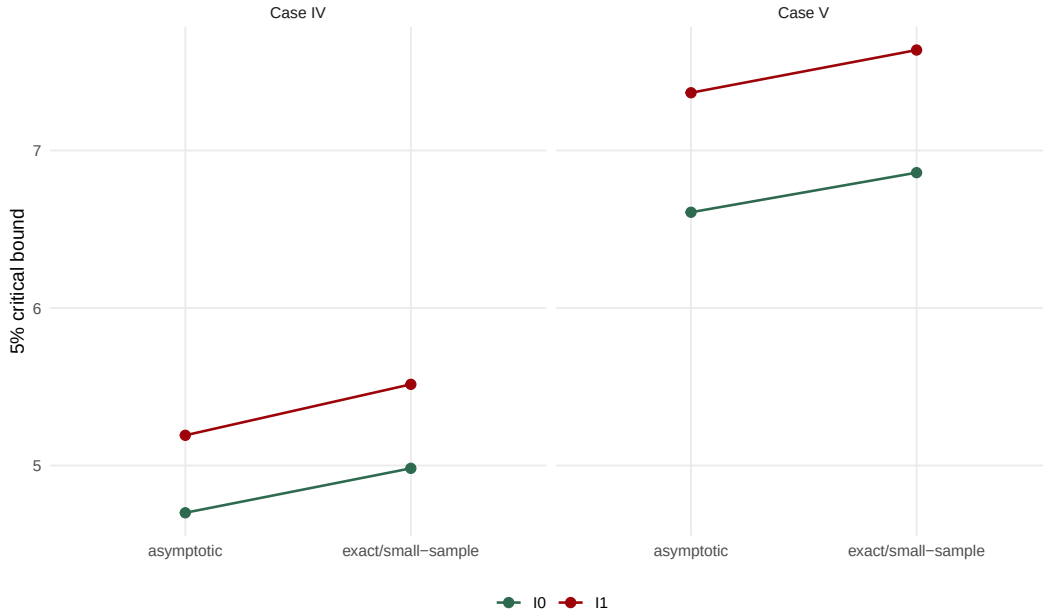


Figure 18: Exact-sample versus asymptotic critical bounds for Cases IV and V ($T = 65$). Exact bounds computed via stochastic simulation (Natsiopoulos and Tzeremes, 2022).

B.9 Dummy Variable Diagnostics

Figure 21 displays squared residuals from the baseline ARDL specification with impulse dummies D_{56} , D_{74} , D_{80} overlaid. The dummies absorb outlier-driven residual spikes identified via squared-error diagnostics (Patterson, 2000).

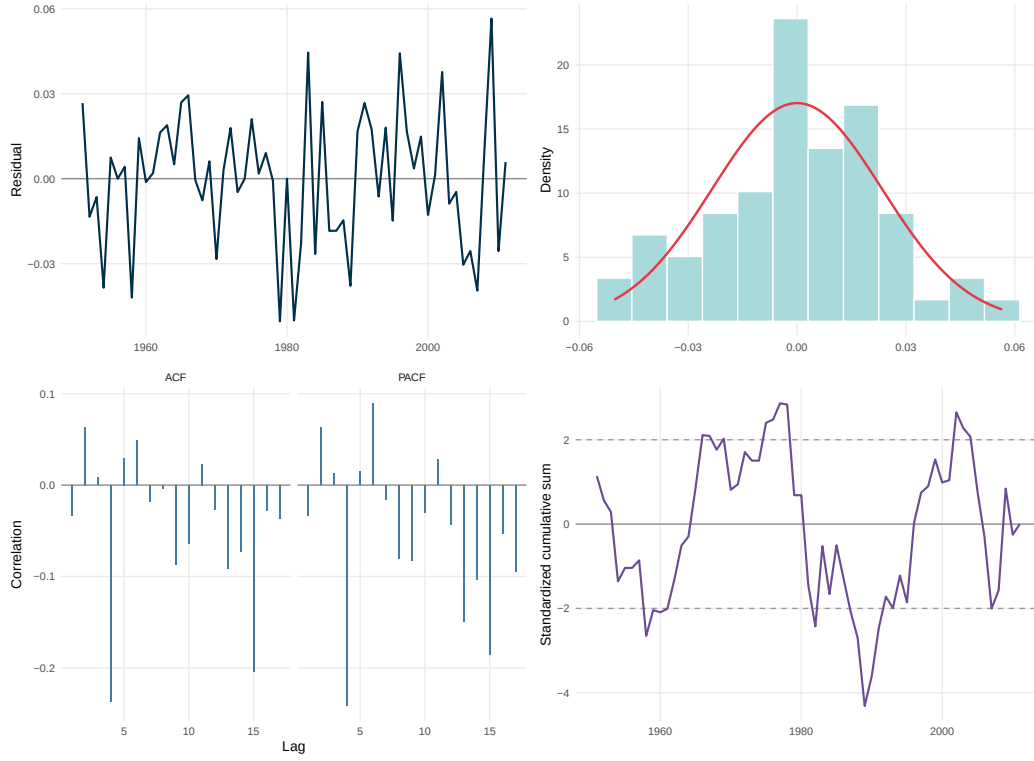


Figure 19: Residual diagnostics from baseline ARDL(2,4): (a) residuals over time, (b) density with normal curve, (c) ACF/PACF, (d) CUSUM test.

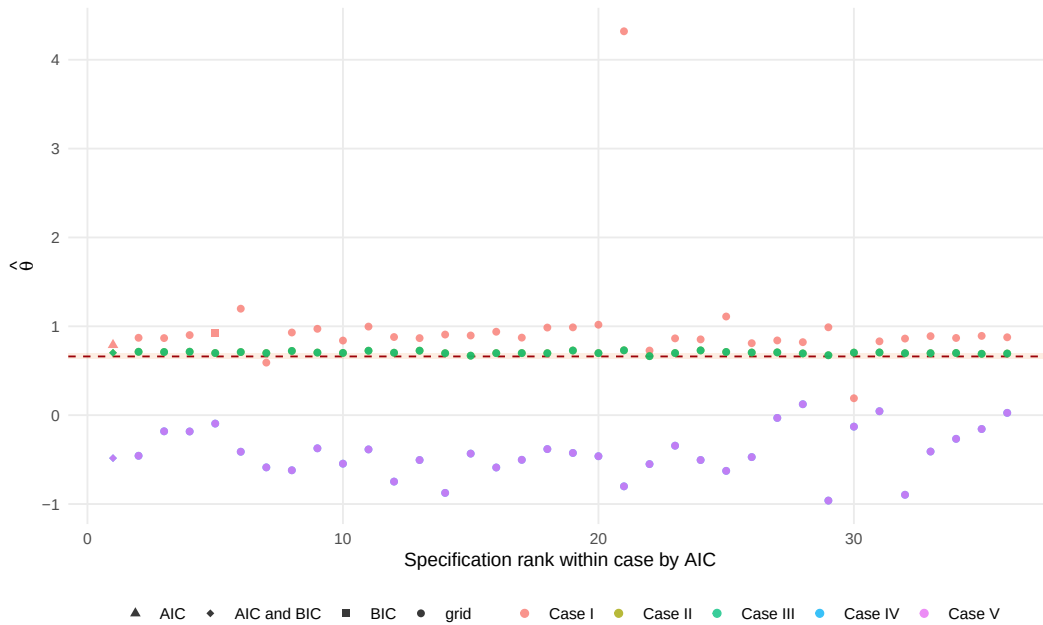


Figure 20: Long-run coefficient $\hat{\theta}$ estimates ranked by AIC across specification grid. Horizontal band marks Shaikh's $0.6609 \pm 5\%$.

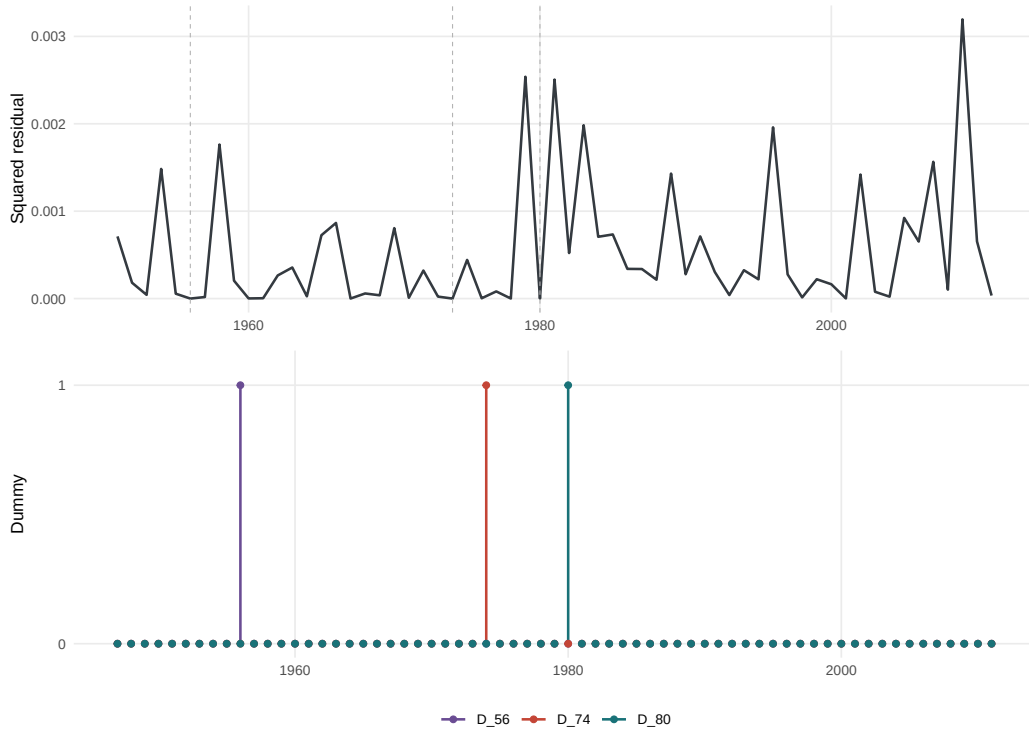


Figure 21: Squared residuals from baseline ARDL with impulse dummies D_{56} , D_{74} , D_{80} (1947–2011). Dummies absorb outlier-driven spikes.

B.10 Series Construction Protocol

Figure 22 visualizes the pipeline from raw data to fitted capacity utilization ($\hat{\mu}_t$). The flowchart traces corrections, deflation, GPIM recursion, ARDL estimation, and residualization.

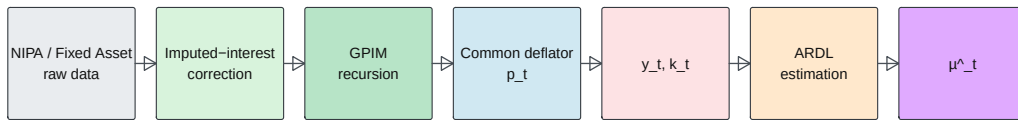


Figure 22: Series construction protocol: raw data $\rightarrow$ corrected output/GPIM capital $\rightarrow$ common deflator $\rightarrow$ ARDL estimation $\rightarrow \hat{\mu}_t$.